Chapter 4

Questions on Lesson Two

Following AC Circuits

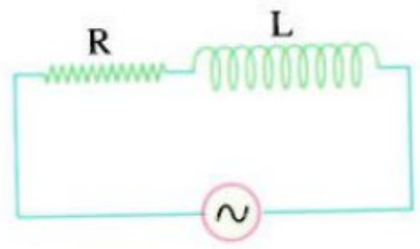
interactive test



The questions signed by ✻ are answered in detail. ● Understand ○ Apply ❖ Higher Order Thinking Skills

Important guidelines

⇒ RL circuit :



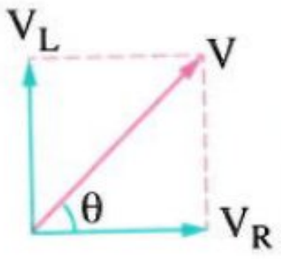
- To determine the total potential difference (V) :

$$V = \sqrt{V_R^2 + V_L^2}$$

- To determine the total impedance (Z) :

$$Z = \sqrt{R^2 + X_L^2}$$

- To determine the phase angle between the current and the total voltage (θ) :



$$\tan\theta = \frac{V_L}{V_R} = \frac{X_L}{R}$$

(Where : (θ) is positive, $90° \geq \theta \geq 0°$)

- To determine the value of the total current (I) :

$$I = \frac{V}{Z} = \frac{V_R}{R} = \frac{V_L}{X_L}$$

- When a DC source is used :

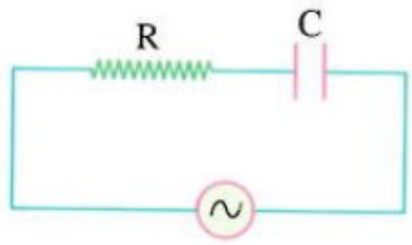
$$I = \frac{V_B}{R} \quad , \quad X_L = 0 \quad , \quad Z = R$$

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⇒ RC circuit :



- To determine the total potential difference (V) :

$$V = \sqrt{V_R^2 + V_C^2}$$

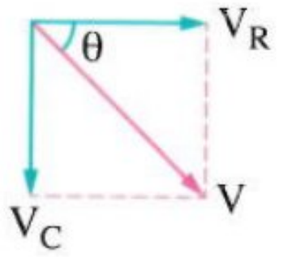
- To determine the total impedance (Z) :

$$Z = \sqrt{R^2 + X_C^2}$$

- To determine the phase angle between the current and the total voltage (θ) :

$$\tan\theta = \frac{-V_C}{V_R} = \frac{-X_C}{R}$$

(Where : (θ) is negative, $-90° \leq \theta \leq 0°$)



- To determine the value of the total current (I) :

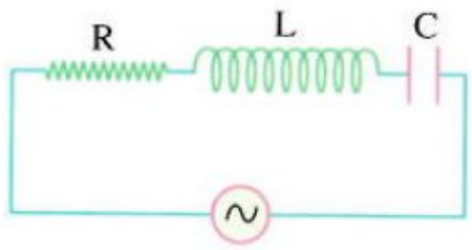
$$I = \frac{V}{Z} = \frac{V_R}{R} = \frac{V_C}{X_C}$$

- In the case of using a DC source, a momentary current passes in the circuit until the capacitor gets charged, then the current gets vanished :

$I = 0$ $\quad X_C = \infty$ $\quad Z = \infty$

⇒ RLC circuit :

- To determine the total potential difference (V) :



$$V = \sqrt{V_R^2 + (V_L - V_C)^2}$$

- To determine the total impedance (Z) :

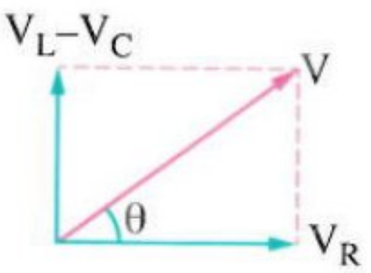
$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

- To determine the phase angle between the current and the total voltage (θ) :

$$\tan\theta = \frac{V_L - V_C}{V_R} = \frac{X_L - X_C}{R}$$

- (θ) is positive when $X_L > X_C$

- (θ) is negative when $X_L < X_C$

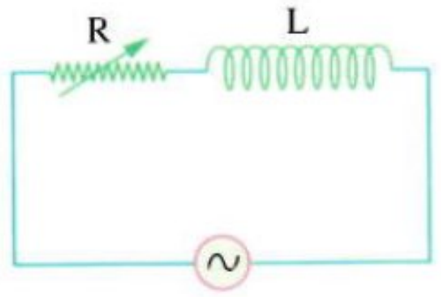


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First Multiple choice questions

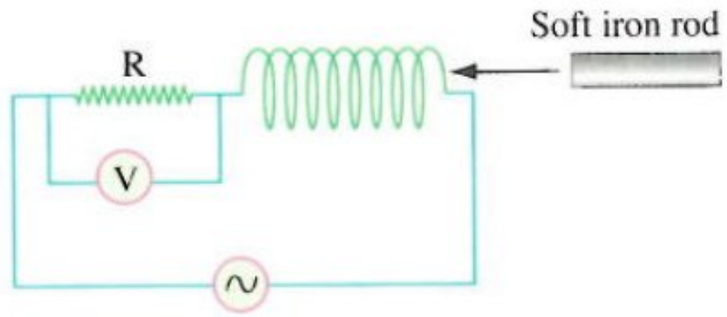
An electric circuit containing an AC source and two components

1 In the opposite electric circuit, if the AC source has a constant effective value of voltage while having a changeable frequency, then the effective value of the electric current that is passing in the circuit increases when ..... .



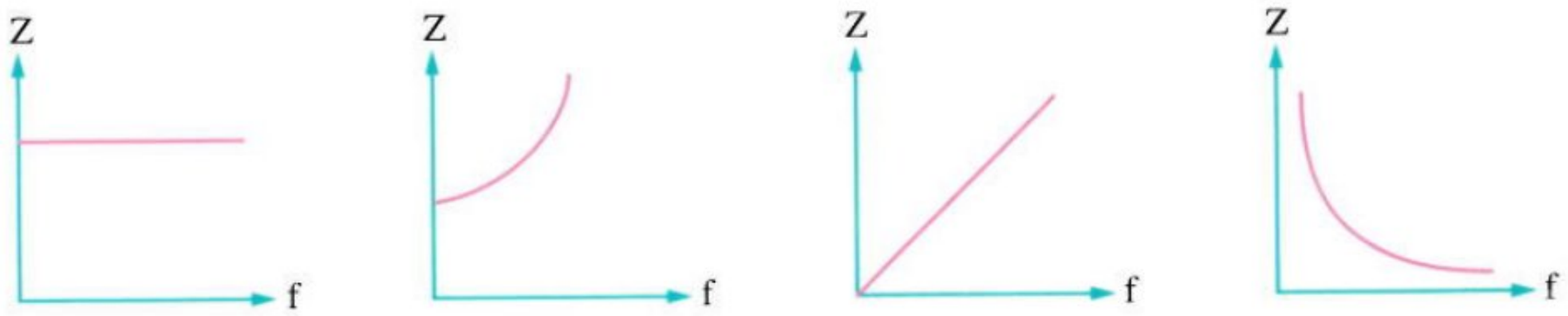
(a) the frequency of the source increases
(b) the frequency of the source decreases
(c) the value of resistor R increases
(d) an iron core is inserted in the coil

2 In the opposite electric circuit, when the iron rod is inserted inside the coil, the reading of the voltmeter ..... .



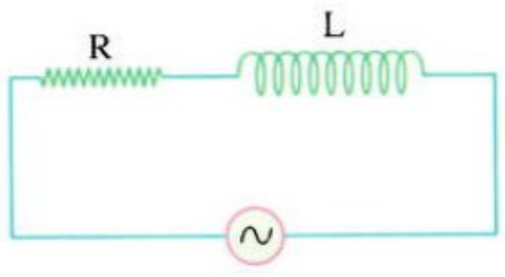
(a) decreases
(b) increases
(c) increases then decreases
(d) remains unchanged

3 An AC circuit consists of an inductor, an ohmic resistor, and an AC source, all connected in series. Which of the following graphs represents the relation between the impedance (Z) of the circuit and the frequency (f) of the circuit current?



(a) (b) (c) (d)

4 In the shown circuit, if the AC power source is replaced by a DC power source whose voltage equals the effective voltage of the AC source, then the ratio of the effective value of the current intensity passing through the circuit in the first case to the current intensity passing through the circuit in the second case is ..... .

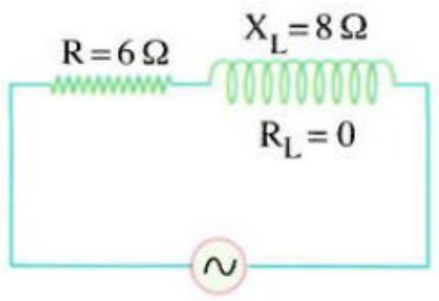


(a) equal to zero (b) less than one (c) equal to one (d) greater than one

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5 In the opposite circuit, the total impedance (Z) equals ..... .



(a) 2 Ω (b) 48 Ω
(c) 14 Ω (d) 10 Ω

6 An inductor has an ohmic resistance of 12 Ω while having an inductive reactance of 18 Ω when a current of frequency f passes through it, then:

(i) Its total impedance in this case is ..... .

(a) 20.1 Ω (b) 16.3 Ω (c) 21.6 Ω (d) 36.2 Ω

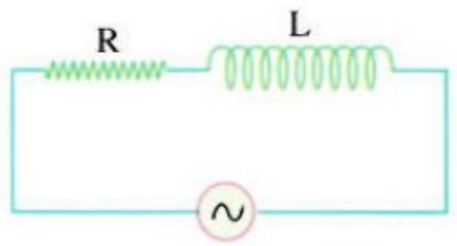
(ii) Its total impedance when the frequency increases to 2 f ..... .

(a) 37.95 Ω (b) 22 Ω (c) 36 Ω (d) 19.99 Ω

7 An electric circuit contains an AC source and an inductor whose inductive reactance is double its ohmic resistance, so the phase angle between the total voltage and the current is ..... .

(a) 26.56° (b) 60° (c) 30.7° (d) 63.4°

8 In the opposite electric circuit, when a current of frequency f passes, $X_L = R$ and the impedance of the circuit equals Z_1, so if the frequency of the current gets increased to 2 f, then the impedance of the circuit Z_2 becomes .....

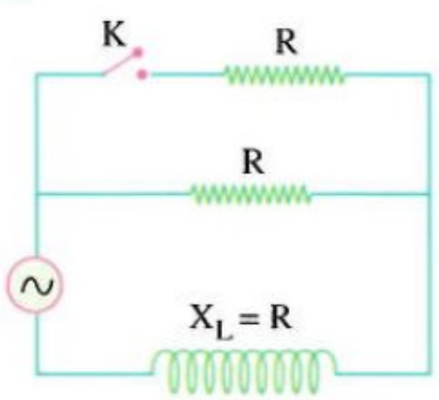


(a) $\frac{Z_1}{2}$ (b) $2\,Z_1$ (c) $1.6\,Z_1$ (d) $2.5\,Z_1$

9 An inductor have an ohmic resistance R and inductive reactance $\sqrt{3}$ R when it is connected to an AC source of frequency f, so the phase angle between the potential difference across the inductor and the current that passes through it equals ..... .

(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{2}$ (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{6}$

10 In the opposite electric circuit, if switch K is closed, the phase angle between the total voltage and the current in the circuit ..... .



(a) decreases by 45° (b) increases by 63.4°
(c) increases by 45° (d) increases by 18.4°

11 An inductor of self-inductance L and resistance 10 Ω is connected to an AC source of potential difference 6.5 V and frequency $\frac{30}{\pi}$ Hz. If the average consumed power in the circuit is $\frac{5}{8}$ W, then the self-inductance (L) of the coil equals ..... .

(a) 1.1 H (b) 0.3 H (c) 0.4 H (d) 0.6 H

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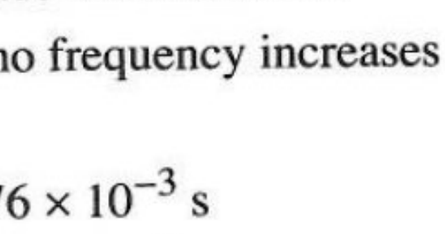
12 An inductor of ohmic resistance 2 Ω is connected to an AC source of effective voltage 6 V, so the inductive reactance of the coil becomes 1 Ω, hence the consumed power in the circuit equals ..... .

(a) 8 W (b) 12 W (c) 14.4 W (d) 18 W

13 An electric bulb of ohmic resistance 44 Ω is connected in series with an inductor of negligible ohmic resistance in an AC circuit. If the frequency of the source is 42 Hz, the effective value of the electromotive force is 220 V and the effective value of the current which passes in the circuit is 4 A, then the self-inductance of the coil equals ..... .

(a) 0.163 H (b) 0.125 H (c) 0.14 H (d) 0.1 H

14 In the opposite AC circuit, when the voltage across the coil is equal to the voltage across the electric resistance, then the frequency of the source equals ..... .



(a) 50 Hz (b) 100 Hz
(c) 200 Hz (d) 400 Hz

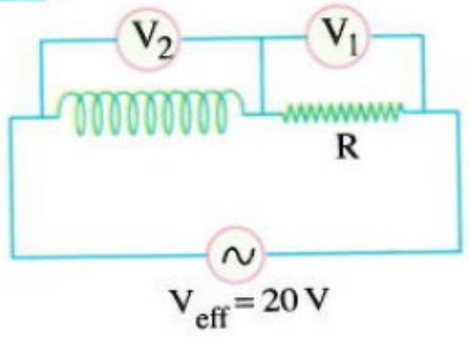
15 * An inductor that has an inductance of 0.01 H and a resistance of 1 Ω is connected to an AC source that has a potential difference of 200 V and a frequency of 50 Hz, then the maximum value of the current ..... .

(a) lags the maximum value of the total voltage by 0.004 s
(b) leads the maximum value of the total voltage by 0.003 s
(c) lags the maximum value of the total voltage by 0.002 s
(d) leads the maximum value of the total voltage by 0.001 s

16 * An AC dynamo of negligible internal resistance is connected in series with an ohmic resistance (R) and an inductor (L) of negligible ohmic resistance. When the dynamo frequency equals 50 Hz, the maximum value of the total voltage leads the maximum value of the current in phase by a time of 2.5×10^{-3} s. If the dynamo frequency increases to 100 Hz, the maximum value of the total voltage will ..... .

(a) lead the maximum value of the current in phase by a time of 1.76×10^{-3} s
(b) lead the maximum value of the current in phase by a time of 3.52×10^{-3} s
(c) lags the maximum value of the current in phase by a time of 1.76×10^{-3} s
(d) lags the maximum value of the current in phase by a time of 3.52×10^{-3} s

17 In the circuit illustrated in the opposite figure:
If the reading of V_1 is 10 V, then the reading of V_2 is ..... .

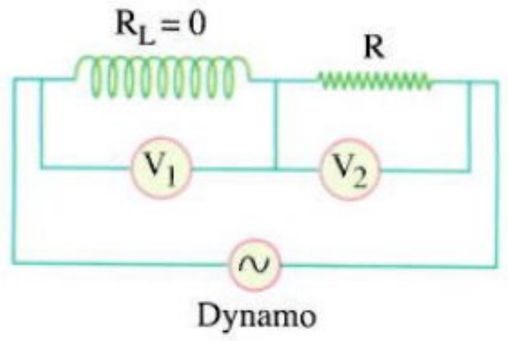


(a) $10\sqrt{3}$ V (b) 10 V
(c) 15 V (d) $10\sqrt{2}$ V

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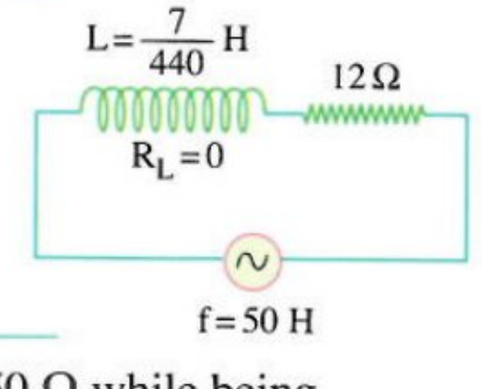
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18 An AC dynamo is connected to an ohmic resistance (R) and an inductor (L) of negligible ohmic resistance, as shown in the opposite electrical circuit. If the readings of the two voltmeters (V_1 and V_2) are equal, which of the following relationships is correct regarding the readings of the two voltmeters after doubling the frequency of the dynamo's rotation?



(a) $V_1 = V_2$ (b) $V_1 > V_2$ (c) $V_1 < V_2$ (d) Indeterminable.

19 ✱ In the opposite figure, the total impedance of the circuit equals ..... .



(a) 17 Ω (b) 13 Ω
(c) 12.2 Ω (d) 7 Ω

20 ✱ An inductor of self inductance $\frac{7}{44}$ H has an inductive reactance 50 Ω while being connected to an AC source, if its ohmic resistance is 30 Ω, then:

(i) The frequency of the current that passes through the inductor equals ..... .

(a) 100 Hz (b) 75 Hz (c) 50 Hz (d) 25 Hz

(ii) The impedance of the inductor equals ..... .

(a) 80 Ω (b) 58.31 Ω (c) 50 Ω (d) 18.75 Ω

21 ✱ The inductive reactance of the operating coil in a circuit regulator is 40 Ω and its resistance is 30 Ω. If the circuit is connected to an AC source of 5 V, then:

(i) The total impedance of the coil is ..... .

(a) 70 Ω (b) 60 Ω (c) 50 Ω (d) 10 Ω

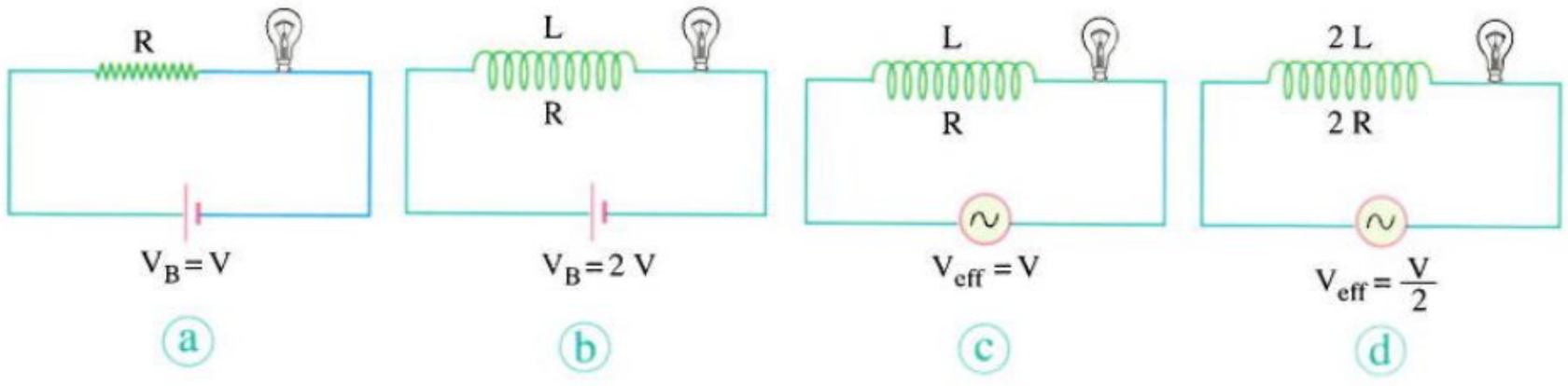
(ii) The electric current passing through the coil is ..... .

(a) 0.07 A (b) 0.09 A (c) 0.1 A (d) 0.2 A

(iii) The phase angle between the total voltage and the current equals ..... .

(a) 36.87° (b) 51.34° (c) 53.13° (d) 59.03°

22 An electric bulb of resistance R is connected in electric circuits with different power sources of zero internal resistances as follows, so in which of the following circuits the brightness of the bulb is highest?



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23 ✳ AC source of emf 100 V and frequency 50 Hz is working in a circuit containing an ohmic resistance of 30 Ω and an inductor of zero resistance and self-inductance $\frac{7}{35}$ H. If they are connected in series, then:

(i) The value of the current passing in the circuit equals ..... .

(a) 1.44 A (b) 1.37 A (c) 1.07 A (d) 0.5 A

(ii) The phase angle between the total voltage and the current equals ..... .

(a) 72.1° (b) 64.49° (c) 59.2° (d) 51.42°

(iii) The potential differences across the resistance (V_R) and the inductor (V_L) respectively are ..... , ...

(a) 50 V, 50 V (b) 20 V, 80 V (c) 43.2 V, 56.8 V (d) 43.2 V, 90.52 V

24 ✳ A resistor of 15 Ω is connected in series with an inductor of zero resistance and an AC source of emf 60 V whose internal resistance is negligible. If the potential difference across the resistor is 45 V, then:

(i) The inductive reactance of the inductor equals ..... .

(a) 11.25 Ω (b) 13.23 Ω (c) 15 Ω (d) 30 Ω

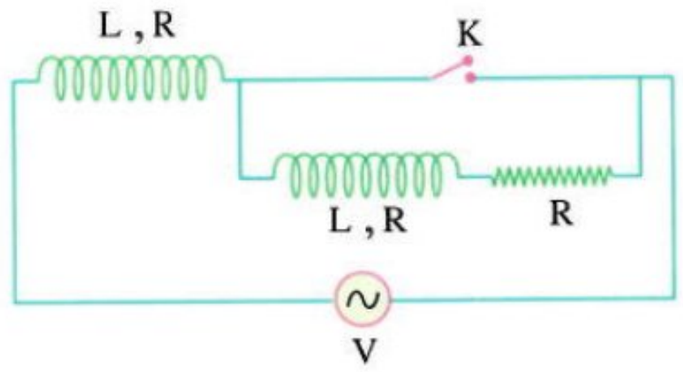
(ii) The potential difference across the inductor terminals equals ..... .

(a) 15 V (b) 24.28 V (c) 39.69 V (d) 41.3 V

25 AC circuit contains an ohmic resistance X and an inductor of inductive reactance X connected in series, so the phase angle between the total voltage and the current equals ..... .

(a) 30° (b) 45° (c) 0° (d) 90°

26 ✳ Two similar coils having the same self-inductance L and the same ohmic resistance R, are connected together with an AC source of voltage V, switch K and an ohmic resistor R as in the opposite figure. When the switch (K) is closed, the effective value of the current and the phase angle between the total voltage and current respectively ..... .



(a) increases, increases (b) increases, decreases

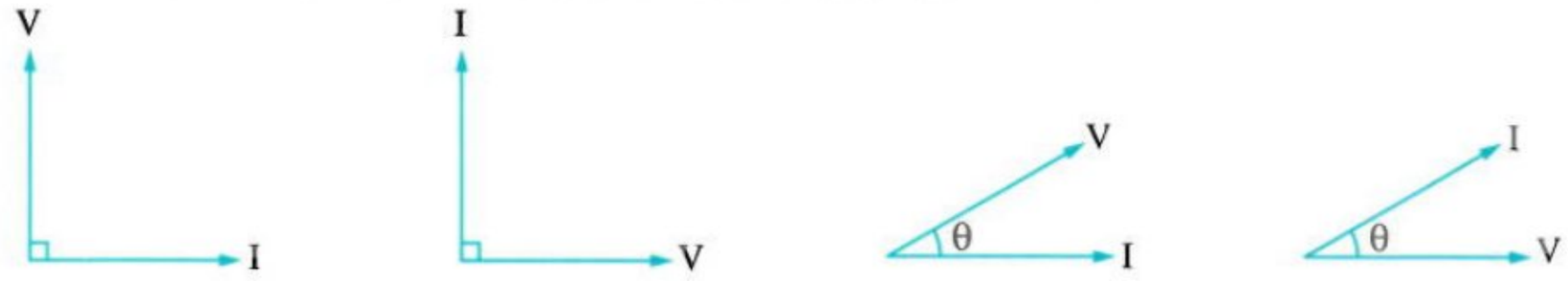
(c) decreases, increases (d) decreases, decreases

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27 An electrical circuit consists of an AC source and an inductor of resistance R, so the shape that represents the phase difference between the phasors of the voltage (V) across the inductor and the current (I) flowing through it is ..... .



(a) (b) (c) (d)

28 Two identical inductors (x and y) are connected together in an electric circuit as shown in figure (1), while figure (2) represents the phase difference between the total voltage phasor (V) and the current phasor (I) in this circuit:

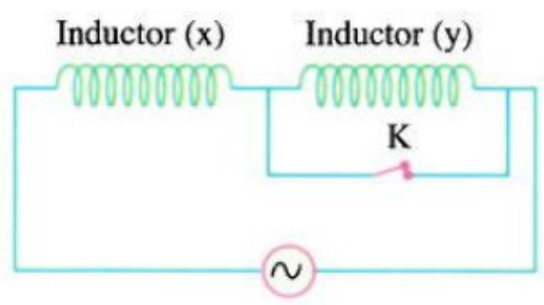


Figure (1)

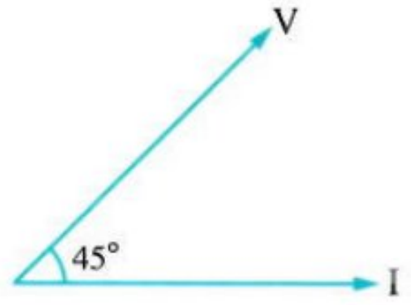
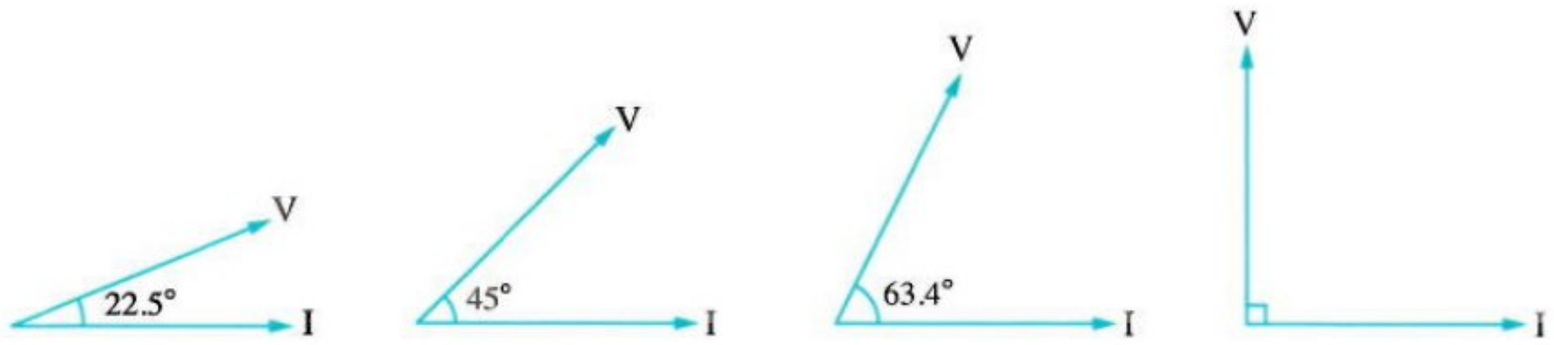


Figure (2)

Which of the following figures represents the phase difference between the total voltage phasor (V) and the current phasor (I) in the circuit after opening switch K?



(a) (b) (c) (d)

29 * An inductor of negligible ohmic resistance is connected in series with an AC power source of emf 260 V and a hot wire ammeter that indicates 2 A, if the ratio between the voltage across the ammeter and across the inductor is $\frac{5}{12}$, then:

(i) The ratio between the resistance of the ammeter and the inductive reactance of the inductor is ..... .

(a) $\frac{5}{12}$ (b) $\frac{12}{5}$ (c) $\frac{1}{2}$ (d) $\frac{2}{1}$

(ii) The impedance of the circuit equals ..... .

(a) 138 Ω (b) 130 Ω (c) 10 Ω (d) 24 Ω

(iii) The resistance of the hot wire ammeter equals ..... .

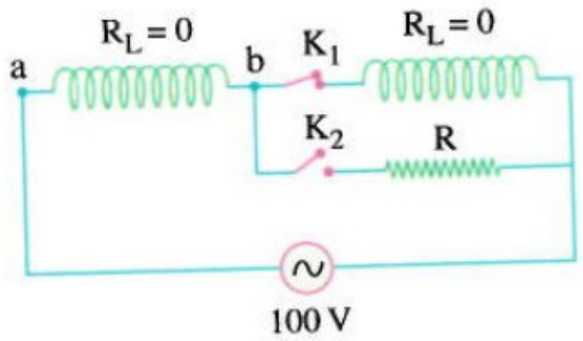
(a) 130 Ω (b) 120 Ω (c) 100 Ω (d) 50 Ω

(iv) The inductive reactance of the inductor equals ..... .

(a) 130 Ω (b) 120 Ω (c) 100 Ω (d) 50 Ω

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30 In the opposite electric circuit, the value of the current flowing through the circuit is 1 A, and the potential difference between points a and b is 80 V. If after opening switch K_1 and closing switch K_2, the potential difference between points a and b remains equal to 80 V, the value of the ohmic resistance R equals ..... .



(a) 40 Ω (b) 50 Ω (c) 60 Ω (d) 80 Ω

31 ✱ A coil is connected to a DC source of emf 11 V, so a current of intensity 2.2 A passes through it and when it is connected to an AC source of frequency 50 Hz and emf 13 V, a current of intensity 1 A passes through it, then the inductance of the coil is equal to ..... .

(a) 0.01 H (b) 0.025 H (c) 0.03 H (d) 0.038 H

32 ✱ An inductor of self inductance $\frac{7}{275}$ H and ohmic resistance 6 Ω is connected to a source of emf 6 V whose internal resistance is negligible, so if :

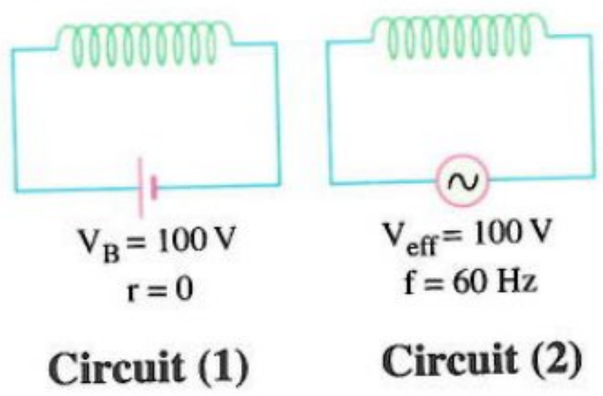
(i) the source is an AC source of frequency 50 Hz, the value of the current through the inductor equals ..... .

(a) 0.2 A (b) 0.42 A (c) 0.6 A (d) 1 A

(ii) the source is a DC source, then the value of the current through the inductor equals ..... .

(a) 0 (b) 0.42 A (c) 0.6 A (d) 1 A

33 An inductor of self-inductance 0.1 H was connected to a DC source of electromotive force 100 V, as shown in circuit (1) so that the electrical power consumed in the inductor was 100 W. Then, it was connected a second time to an AC source of effective voltage 100 V and frequency 60 Hz, as shown in circuit (2). So, with neglecting the internal resistance of the two sources, the electrical power consumed in the inductor ..... .



Circuit (1) **Circuit (2)**

(a) remains 100 W (b) becomes 187.5 W
(c) becomes 87.5 W (d) becomes 12.5 W

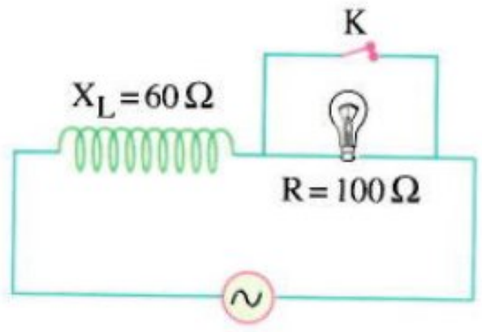
34 ✱ An inductor of self inductance 2 H is connected in series to a resistor of 1950 Ω and an AC source of frequency $\frac{500}{\pi}$ Hz, so the phase angle between the current and the total voltage becomes 45°, then the ohmic resistance of the inductor is equal to ..... .

(a) 500 Ω (b) 250 Ω (c) 150 Ω (d) 50 Ω

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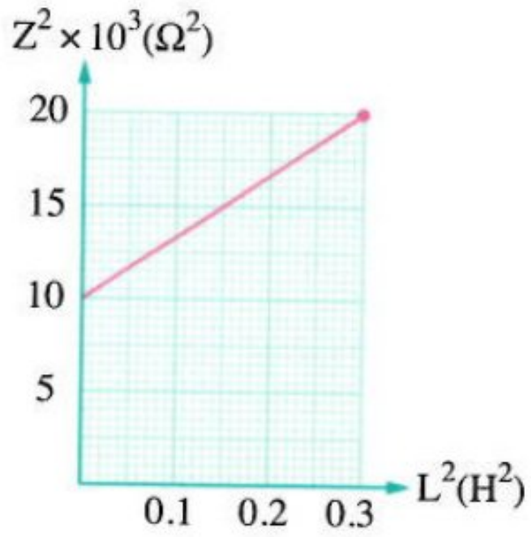
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35 ✻ In the opposite electric circuit, an AC source of voltage 200 V is connected to a lamp of resistance 100 Ω and an inductor to have an inductive reactance of 60 Ω where an electric current of 2 A flows through the circuit, so the electrical power consumed in the lamp after opening switch K is approximately equal to ..... .



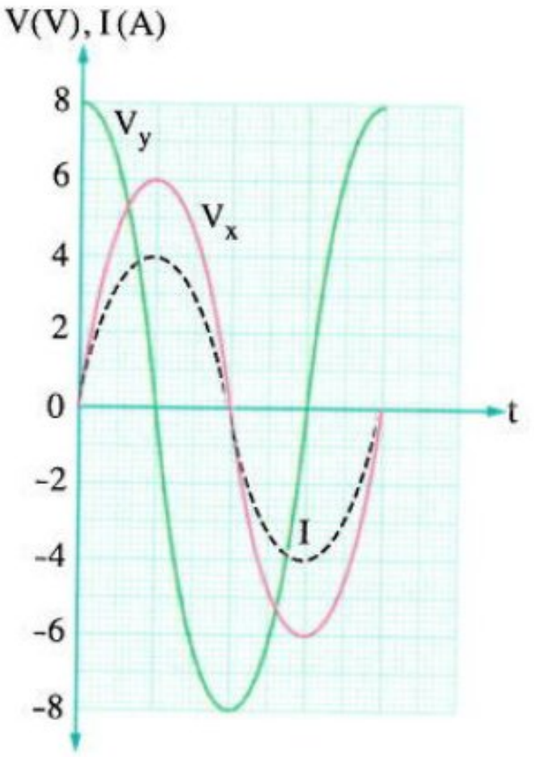
(a) 111 W (b) 222 W (c) 333 W (d) 444 W

36 ✻ An AC circuit consists of an AC source of angular frequency ω, an ohmic resistance (R), and an inductor of negligible ohmic resistance whose self-inductance can be changed. The opposite graph represents the relationship between the square of the circuit impedance (Z^2) and the square of the inductor self-inductance (L^2), so ..... .



	The value of the ohmic resistance (R)	The angular frequency of the source (ω)
(a)	$10^2\ \Omega$	29.05 rad/s
(b)	$10^2\ \Omega$	182.6 rad/s
(c)	$10^4\ \Omega$	29.05 rad/s
(d)	$10^4\ \Omega$	182.6 rad/s

37 ✻ An AC circuit contains an AC source and two pure components x, y connected in series and the opposite graph shows the change of each of the voltages (V_x), (V_y) in volts and the current (I) in amperes versus time (t), then:



(i) The two components x, y respectively are ..... .

(a) an ohmic resistance and an inductor
(b) an inductor and an ohmic resistance
(c) an ohmic resistance and a capacitor
(d) a capacitor and an inductor

(ii) The phase angle between the total voltage and the current equals ..... .

(a) 26.56° (b) 33.69° (c) 36.86° (d) 53.13°

(iii) The electromotive force of the source equals ..... .

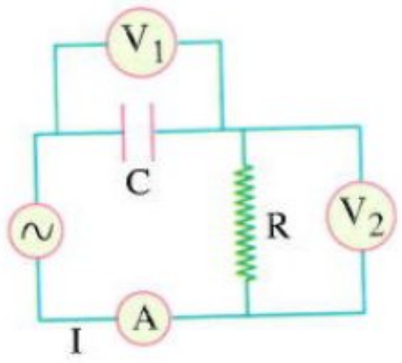
(a) 6 V (b) 7.07 V (c) 8 V (d) 14 V

(iv) The impedance of the circuit equals ..... .

(a) 1 Ω (b) 2 Ω (c) 2.5 Ω (d) 3.5 Ω

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38 The opposite figure shows an AC circuit that contains a capacitor C and a resistor R, so which of the following choices is correct?



(a) Voltage V_2 and current I are in the same phase.
(b) Voltage V_1 leads voltage V_2 in phase.
(c) Voltage V_1 and current I are in the same phase.
(d) Voltages V_1, V_2 and current I are in the same phase.

39 The impedance of an AC circuit that contains a capacitor and a resistor is ..... .
(a) equal to the difference between the capacitive reactance of the capacitor and the resistance
(b) zero
(c) equal to the algebraic sum of the capacitive reactance and the resistance
(d) less than the algebraic sum of the capacitive reactance of the capacitor and the resistance

40 The value of the current in an AC circuit that contains a capacitor of variable capacitance and rheostat connected in series increases by ..... .
(a) increasing the capacitance of the capacitor
(b) decreasing the capacitance of the capacitor
(c) increasing the resistance of the rheostat
(d) decreasing the frequency of the AC source

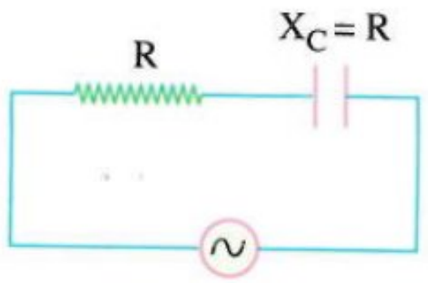
41 A capacitor of capacitance C and an ohmic resistance R are connected in series with an AC dynamo where the capacitive reactance equals the resistance ($X_C = R$), if the frequency of dynamo's rotation decreases, then the relation between the potential difference across the two plates of the capacitor and the voltage across the resistance terminals becomes .....
(a) $V_R > V_C$ (b) $V_C > V_R$ (c) $V_R = V_C = 0$ (d) $V_R = V_C \neq 0$

42 An AC source of frequency $\frac{50}{\pi}$ Hz is connected to an electric circuit that contains an ohmic resistance of 1 kΩ and another component, if the phase difference between the total voltage and the current in the circuit is $\frac{\pi}{4}$, then the other component connected in the circuit is ..... .
(a) a capacitor of capacitance 10 μF
(b) a capacitor of capacitance 1 μF
(c) an inductor of self-inductance 5 H
(d) an inductor of self-inductance 1 H

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43 In the opposite electrical circuit, if the value of the capacitive reactance of the capacitor is equal to the value of the ohmic resistance (R), and the instantaneous value of the voltage across the source is given by the relation; $(V = 20 \sin(\omega t))$, then the instantaneous value of the current through the circuit is given by the relation; .....



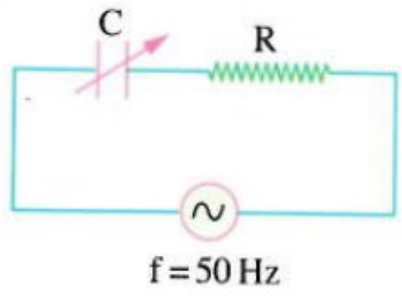
(a) $I = \frac{20}{R} \sin(\omega t + \frac{\pi}{4})$

(b) $I = \frac{10\sqrt{2}}{R} \sin(\omega t + \frac{\pi}{6})$

(c) $I = \frac{10\sqrt{2}}{R} \sin(\omega t - \frac{\pi}{4})$

(d) $I = \frac{10\sqrt{2}}{R} \sin(\omega t + \frac{\pi}{4})$

44 * In the opposite electric circuit, the phase angle between the current and the total voltage becomes 30° when the capacitance of the capacitor equals C_1 and becomes 60° when the capacitance of the capacitor changes to C_2, then .....



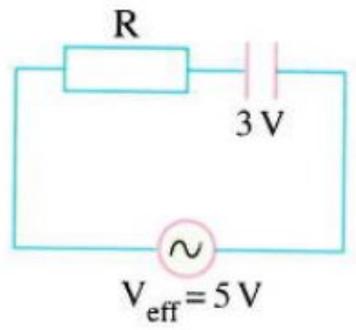
(a) $C_2 = \frac{C_1}{3}$

(b) $C_2 = \frac{2C_1}{3}$

(c) $C_2 = \frac{2C_1}{5}$

(d) $C_2 = \frac{3C_1}{5}$

45 In the AC circuit illustrated in the figure, if the potential difference across capacitor C is 3 V, then the potential difference across resistor R equals ..... .



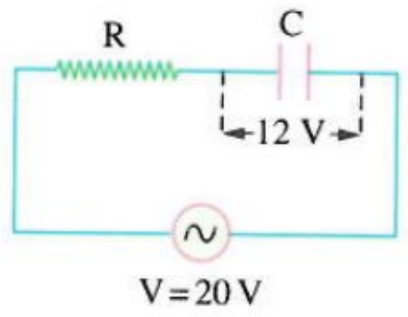
(a) 1 V

(b) 2 V

(c) 3 V

(d) 4 V

46 In the shown circuit, if the effective value of the electric current through the circuit is 2 A, then the resistance of resistor R equals ..... .



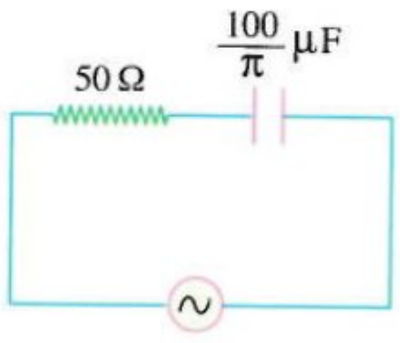
(a) 4 Ω

(b) 6 Ω

(c) 8 Ω

(d) 12 Ω

47 The opposite diagram represents an AC circuit, so when the potential difference across the capacitor becomes equal to the potential difference across the ohmic resistance, the frequency of the source equals ..... .



(a) 50 Hz

(b) 60 Hz

(c) 100 Hz

(d) 500 Hz

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48 When an AC current of frequency 500 Hz is set up through a capacitor of capacitance $\frac{7}{22}$ μF that is connected to a non-inductive ohmic resistance of 1000 Ω, then:

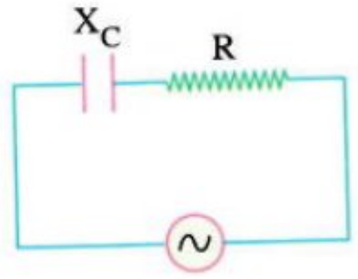
(i) The total impedance is ..... .

(a) 1414.2 Ω (b) 2000 Ω (c) 318.2 Ω (d) 5×10^4 Ω

(ii) The total voltage lags the current by a phase angle ..... .

(a) 50° (b) 45° (c) 90° (d) 63.75°

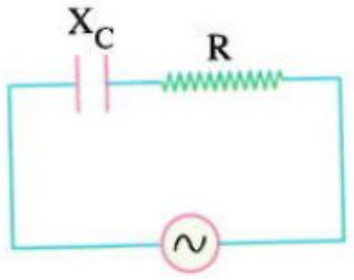
49 In the opposite circuit, if the capacitive reactance X_C equals three times the ohmic resistance R, then the impedance (Z) equals ..... .



(a) $\sqrt{2}$ R (b) R

(c) $\sqrt{10}$ R (d) 4 R

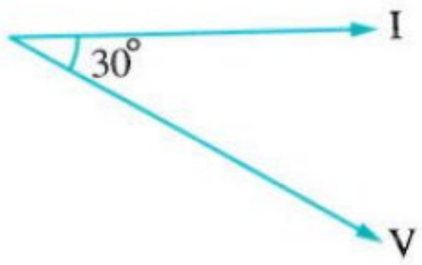
50 * In the opposite circuit, when a current of frequency f passes in the circuit, X_C = R, then if the frequency is increased to 2 f, the impedance ..... .



(a) increases to its double (b) decreases to its half

(c) becomes 1.1 R (d) becomes 0.5 R

51 The opposite figure represents the phasor diagram of the current (I) and the total voltage (V) in an AC circuit containing two pure components x, y and the AC source, then the two components x, y could be ..... .



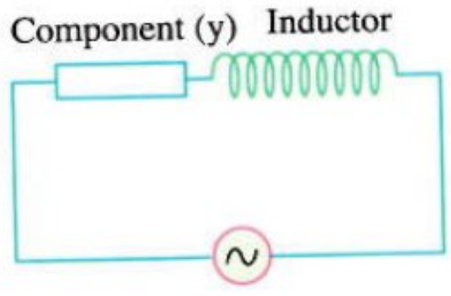
(a) an ohmic resistance and a capacitor

(b) an ohmic resistance and a coil

(c) a coil and a capacitor

(d) an ohmic resistance and an ohmic resistance

52 An inductor of negligible ohmic resistance is connected to an unknown component (y) and an AC power source as in the opposite figure, if the total voltage = The potential difference across the inductor terminals + The potential difference across the terminals of component (y), so component (y) should be ..... .



(a) an ohmic resistance (b) an inductor of negligible ohmic resistance

(c) a capacitor (d) an inductor having an ohmic resistance

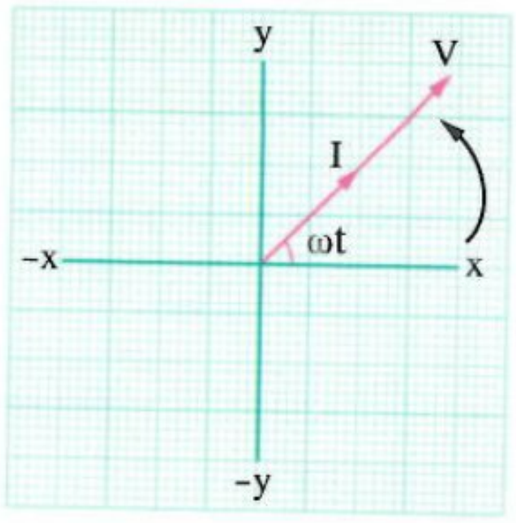
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53 The opposite figure represents the phasors of the total voltage (V) and current (I) in an AC circuit consisting of AC source and two pure components (a, b), then the components (a, b) are ..... .



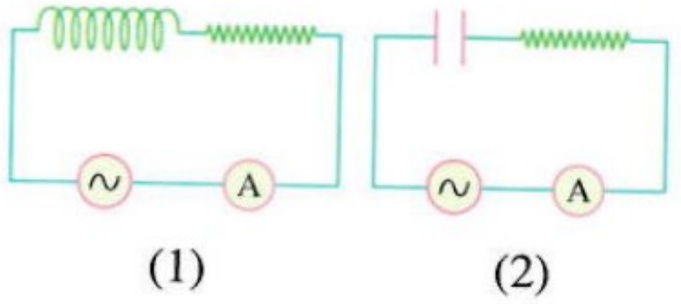
(a) an ohmic resistance and an inductor

(b) an ohmic resistance and capacitor

(c) two ohmic resistances

(d) an inductor and a capacitor

54 An AC source, a hot-wire ammeter, and an ohmic resistor were connected in series once with an inductor of negligible ohmic resistance, as shown in circuit (1), and another time with a capacitor, as shown in circuit (2). If the reading of the hot-wire ammeter is the same in both circuits, which of the following statements is **incorrect**?



(1) (2)

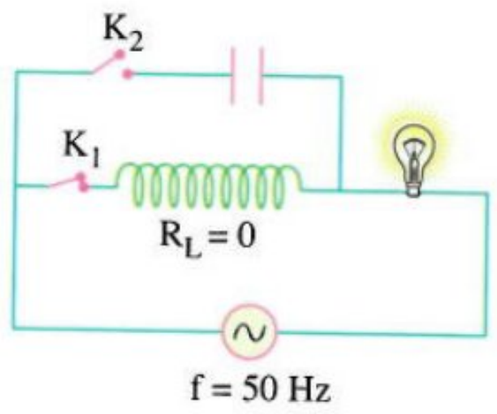
(a) The inductive reactance of the inductor = The capacitive reactance of the capacitor

(b) The voltage across the inductor = The voltage across the capacitor

(c) The current lags the total voltage by an angle θ in both circuits.

(d) The impedances of the two circuits are equal.

55 An AC source with a frequency of 50 Hz is connected to an inductor of self-inductance $\frac{1}{\pi}$ H and negligible internal resistance, a capacitor of capacitance $\frac{100}{\pi}$ μF, and a lamp, as shown in the opposite electrical circuit. What happens to the brightness of the lamp after opening switch K_1 and closing switch K_2?



(a) Increases. (b) Decreases but doesn't vanish.

(c) Does not change. (d) The answer cannot be determined.

56 * A resistor of 300 Ω is connected in series with a capacitor of capacitive reactance 265 Ω and an AC power source of frequency 100 Hz, if the voltage across the capacitor was 5 V, then:

(i) The capacitance of the capacitor equals ..... .

(a) 0.42 μF (b) 0.9 μF (c) 3 μF (d) 6 μF

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(ii) The value of the electric current in the circuit approximately equals ..... .

(a) 0.008 A (b) 0.017 A (c) 0.019 A (d) 0.01 A

(iii) The potential difference between the terminals of the resistor equals ..... .

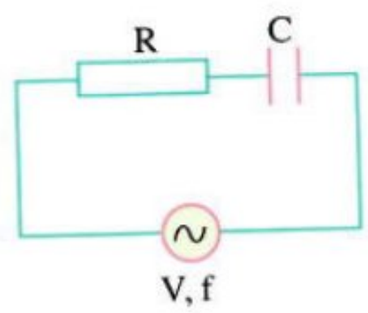
(a) 4.7 V (b) 5 V (c) 5.7 V (d) 6 V

57 ✳ An AC source of emf 200 V and frequency 50 Hz is connected to a lamp (25 W, 100 V) and a capacitor of $\frac{100}{3\pi}$ μF, then the filament of the lamp carries an electric current that equals ..... .

(a) 0.15 A and it doesn't glow (b) 0.2 A and it glows

(c) 0.4 A and it burns out (d) 0.45 A and it burns out

58 ✳ In the circuit illustrated in the figure:



If the value of the phase angle between the current and the total voltage is 45°, then the phase angle between them when:

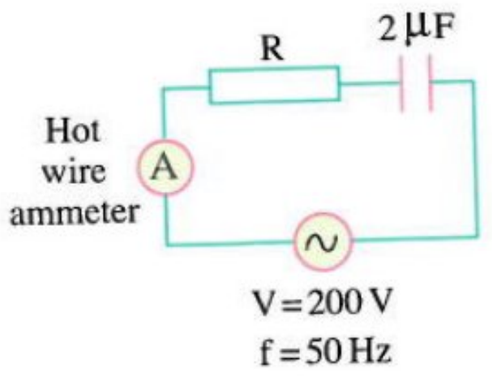
(i) the capacitor is connected in series to another capacitor of capacitance C becomes ..... .

(a) 26.57° (b) – 63.4° (c) 63.4° (d) – 26.57°

(ii) the resistor is connected in series to another resistor of resistance R becomes ..... .

(a) 26.57° (b) – 63.4° (c) – 26.57° (d) 63.4°

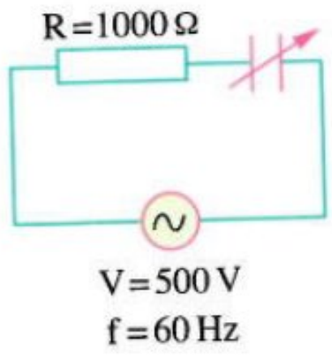
59 ✳ From the circuit illustrated in the figure, if the effective value of the current passing in the circuit is 0.02 A, the resistance of R equals ..... .



(a) 11590.91 Ω (b) 10000 Ω

(c) 9872.64 Ω (d) 8409.81 Ω

60 ✳ From the circuit illustrated in the figure, the capacitance of the capacitor when:



(i) the effective value of the passing electric current becomes 0.25 A, equals ..... .

(a) 1.53 μF (b) 2.65 μF

(c) 3.42 μF (d) 4.59 μF

(ii) the phase angle between the current and the total voltage becomes – 45° equals ..... .

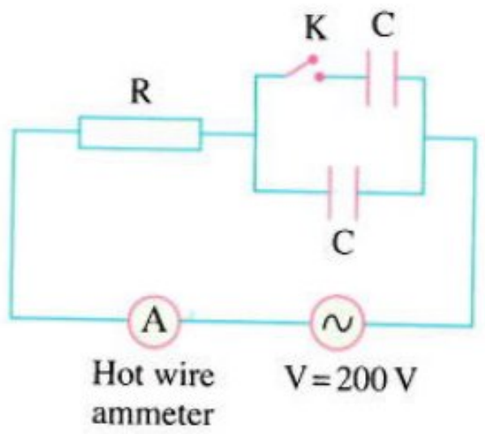
(a) 5.3 μF (b) 4.33 μF (c) 2.65 μF (d) 1.42 μF

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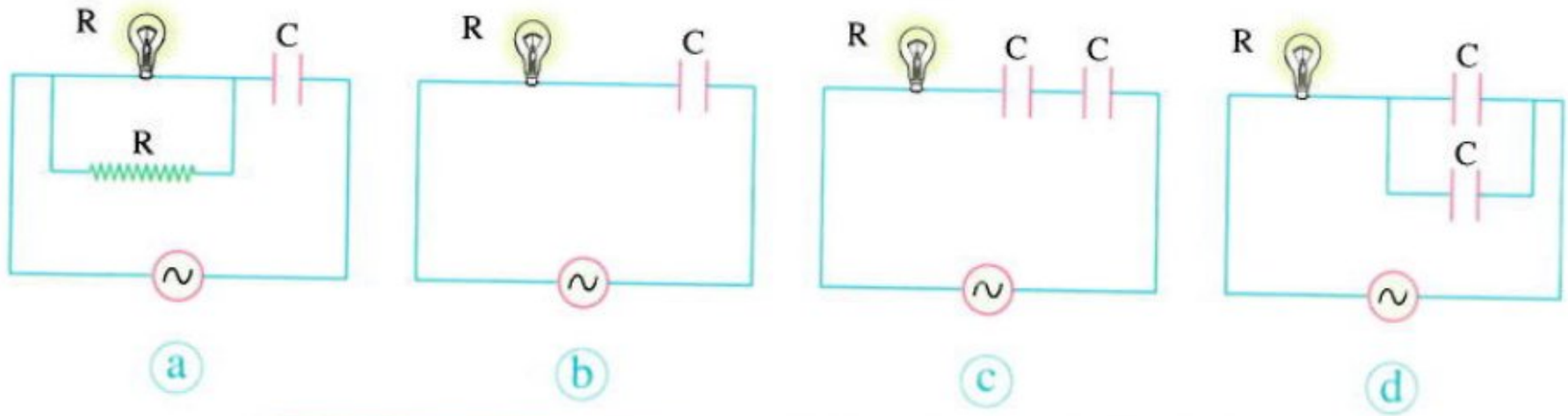
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61 * In the circuit illustrated in the figure, the total ohmic resistance is 500 Ω. If the reading of the hot wire ammeter when opening switch K is 0.2 A, then its reading when closing switch K is ..... .



(a) 0.4 A
(b) 0.3 A
(c) 0.27 A
(d) 0.2 A

62 In the following electrical circuits, if all the electrical sources are identical, and the capacitance of each of the capacitors is C and the resistance of each of the lamps is R, in which of these circuits is the brightness of the lamp highest?



(a) (b) (c) (d)

63 An AC circuit contains an inductor L of zero resistance and a capacitor C. If they are connected in series, so the voltage V_L ..... .

(a) leads V_C in phase by 90°
(b) lags V_C in phase by 90°
(c) is in phase with V_C
(d) leads V_C in phase by 180°

64 An AC, LC circuit consists of a source whose frequency can be changed, a capacitor, and an inductor of negligible ohmic resistance connected in series. When the source frequency is f, the capacitive reactance (X_C) of the capacitor becomes 500 Ω and the inductive reactance (X_L) of the inductor becomes 100 Ω, so the magnitude of change in the phase angle between the total voltage in the circuit and the current after:

(i) increasing the source frequency to 2f will be ..... .

(a) 0° (b) 45° (c) 90° (d) 180°

(ii) increasing the source frequency to 5f will be ..... .

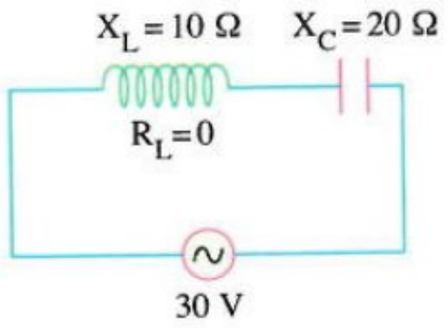
(a) 0° (b) 45° (c) 90° (d) 180°

65 In an AC circuit whose components are in series, it was found that the value of current intensity at a certain instant equals zero when the potential difference across the source is maximum value, so the AC source might be connected with ..... .

(a) an inductor of negligible ohmic resistance
(b) a capacitor
(c) an inductor of zero ohmic resistance and a capacitor
(d) all the previous

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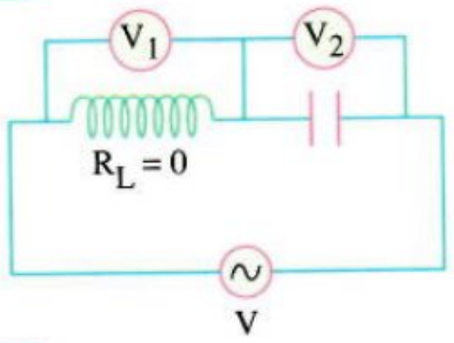
66 In the opposite circuit, the value of the current that passes through the circuit is ..... .



(a) 1 A (b) $\frac{3}{\sqrt{5}}$ A

(c) 2 A (d) 3 A

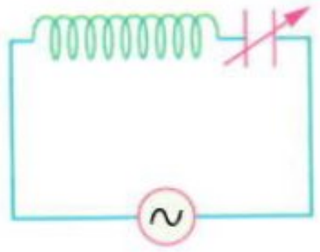
67 In the opposite circuit, if $V_1 = V$, so the reading of V_2 equals ..... .



(a) V (b) 2 V

(c) $\sqrt{2}$ V (d) 3 V

68 * In the shown circuit, a capacitor of variable capacitance that has a capacitive reactance $(X_C)_1$ and an inductor of negligible ohmic resistance that has an inductive reactance X_L are connected in series to an AC source such that $X_L > (X_C)_1$ and the effective value of the electric current is I, then when the capacitance of the capacitor is decreased to its quarter, $(X_C)_2 > X_L$ and the effective value of the electric current gets increased to the double, so the ratio $\frac{X_L}{(X_C)_1}$ is ..... .

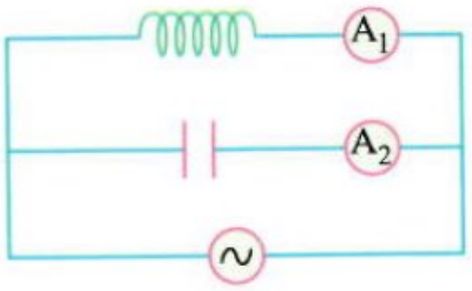


(a) $\frac{1}{3}$ (b) $\frac{2}{3}$ (c) $\frac{3}{1}$ (d) $\frac{3}{5}$

69 An AC source of frequency f is connected to a capacitor and an inductor of negligible ohmic resistance so that the capacitive reactance of the capacitor becomes double the inductive reactance of the inductor, so if the frequency of the source is increased to the double, then the capacitive reactance of the capacitor will be ..... the inductive reactance of the inductor.

(a) double (b) equal to (c) half (d) four times

70 When replacing the source in the opposite electric circuit with another of the same voltage and higher frequency, which of the following choices will be correct?



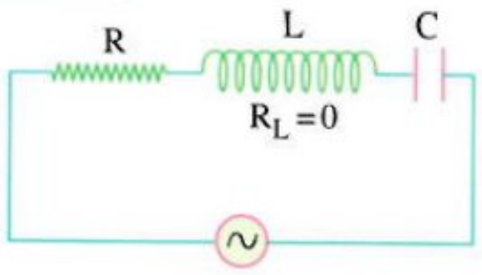
	The hot wire ammeter reading (A_1)	The hot wire ammeter reading (A_2)
(a)	Increases	Decreases
(b)	Decreases	Increases
(c)	Decreases	Decreases
(d)	Increases	Increases

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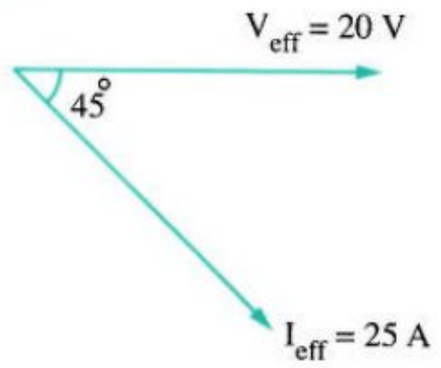
An electric circuit containing an AC source and three components

71 In which of the components shown in the opposite electric circuit is the electrical energy converted into thermal energy?



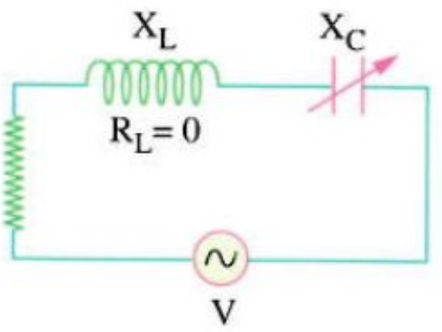
(a) The ohmic resistance (R).
(b) The inductor (L).
(c) The capacitor (C).
(d) All three components.

72 The opposite figure represents the phasor diagram for the potential difference and the current intensity in an AC circuit, so the circuit is ..... .



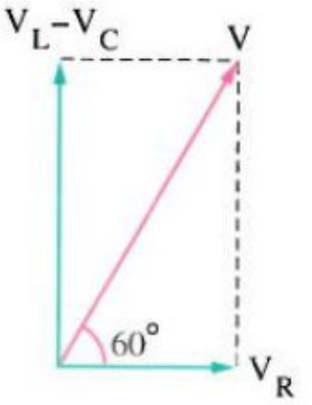
(a) RLC only
(b) RL only
(c) RC only
(d) RL or RLC

73 In the opposite circuit, when $(X_C)_1 = \frac{1}{2}X_L$, the value of the current that passes in the circuit equals I. If the capacitance gets decreased gradually until $(X_C)_2 = \frac{3}{2}X_L$, the passing current intensity through the circuit will ..... .



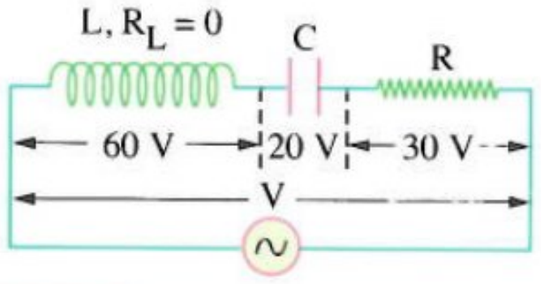
(a) decrease until it vanishes
(b) decrease until it vanishes then increases
(c) increase
(d) increase then decrease till reaching its original value

74 * The opposite figure shows the voltage phasors for an RLC circuit, then the total impedance of the circuit equals ..... .



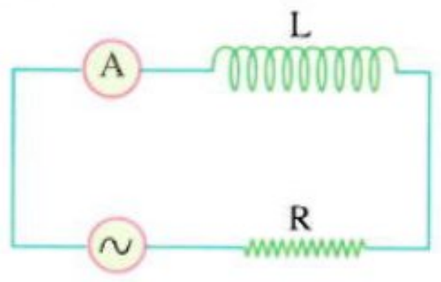
(a) $\frac{R}{2}$
(b) $\frac{3R}{4}$
(c) R
(d) 2 R

75 In the opposite electric circuit, the voltage of the AC source is ..... .



(a) 40 V
(b) 50 V
(c) 70 V
(d) 110 V

76 When adding a capacitor in series in the opposite circuit, it has been noticed that the hot wire ammeter reading did not change. In this case the capacitive reactance of the capacitor is ..... the inductive reactance of the coil.



(a) half
(b) equal to
(c) double
(d) three times as

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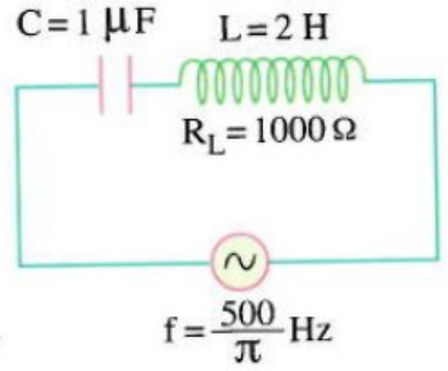
77 AC circuit consists of a resistance R, an inductor L and a capacitor C that are connected in series, if $X_C = 2\,X_L = 2\,R$, then the total potential difference ..... .

(a) leads V_R in phase by 90°
(b) leads V_R in phase by 45°
(c) lags V_R in phase by 90°
(d) lags V_R in phase by 45°

78 Series RLC circuit has a total impedance of 20 Ω, if the value of the ohmic resistance is 10 Ω and the inductive reactance of the coil is larger than the capacitive reactance of the capacitor, so the phase angle between the total voltage and the circuit current equals ..... .

(a) 30°
(b) 45°
(c) 60°
(d) 90°

79 In the shown AC circuit:



(i) The total impedance is ..... .

(a) 1000 Ω
(b) 2000 Ω
(c) 5000 Ω
(d) $1000\sqrt{2}$ Ω

(ii) The phase angle between the total voltage and current is ..... .

(a) 90°
(b) 30°
(c) 45°
(d) 0°

80 If the phase angle between the total voltage and the current in RLC circuit is 45° and ($R = X_C$), then the ratio between the inductive reactance of the coil and the capacitive reactance of the capacitor $\left(\frac{X_L}{X_C}\right)$ equals ..... .

(a) $\frac{2}{1}$
(b) $\frac{\sqrt{2}}{1}$
(c) $\frac{1}{2}$
(d) $\frac{1}{1}$

81 A series circuit consists of an AC source of frequency $\frac{500}{\pi}$ Hz where its effective voltage is 200 V, an inductor of self-inductance 0.08 H, a resistance of 30 Ω and a capacitor. If the total impedance of the circuit is 50 Ω, then the phase angle between the total voltage and the current equals ..... .

(a) 42.19°
(b) 46.12°
(c) 49.17°
(d) 53.13°

82 * AC source of effective voltage 50 V and frequency $\frac{500}{\pi}$ Hz is connected in series to a resistance of 300 Ω, a coil of negligible ohmic resistance which has self inductance 0.9 H and a capacitor of capacitance 2 μF, then:

(i) The circuit impedance equals ..... .

(a) 500 Ω
(b) 806.23 Ω
(c) 1431.78 Ω
(d) 1700 Ω

(ii) The value of electric current passing in the circuit equals ..... .

(a) 0.03 A
(b) 0.1 A
(c) 0.17 A
(d) 0.25 A

(iii) The consumed power in the circuit equals ..... .

(a) 4 W
(b) 3 W
(c) 2 W
(d) 1.47 W

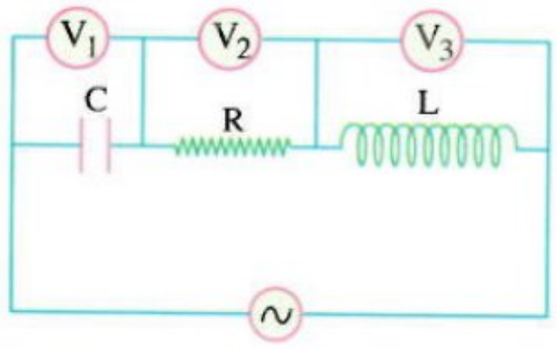
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83 * An electric circuit is composed of an ohmic resistance of 15 Ω, an inductor of self inductance 0.08 H and a capacitor of capacitance 30 μF. They all are connected in series with an AC dynamo of angular speed 500 rad.s^{-1}, then the total voltage ..... .

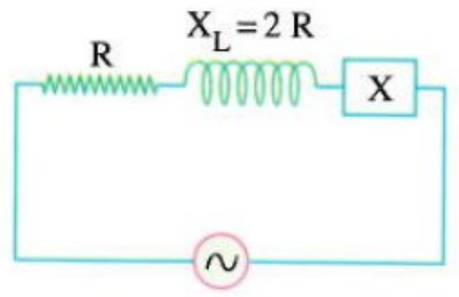
(a) leads the current by an angle of 81.9°
(b) lags the current by an angle of 60.65°
(c) leads the current by an angle of 60.65°
(d) lags the current by an angle of 81.9°

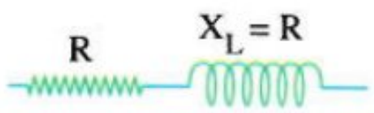
84 An inductor of negligible ohmic resistance, an ohmic resistance and a capacitor are connected together to an AC source in series as shown in the opposite figure, so that the relation between the readings of the voltmeters is $V_2 = V_3 = 2\,V_1$, so when placing a rod of soft iron inside the coil, then the readings of the two voltmeters V_1, V_2 respectively will ..... .



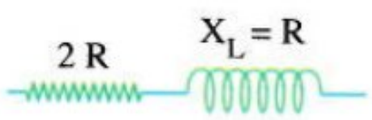
(a) decrease, decrease
(b) decrease, increase
(c) increase, decrease
(d) increase, increase

85 Which of the following choices represents the components that can be placed in position X so that the phase angle between the total voltage and the current becomes 45°?
(Where the ohmic resistances of the coils are negligible)

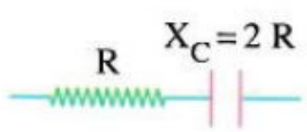




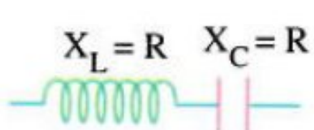
(a)



(b)



(c)



(d)

86 An electric circuit consists of a capacitor of capacitive reactance 80 Ω, an inductor of self inductance 0.28 H and an ohmic resistance in the form of a wire of length 12 m and cross-sectional area 7×10^{-5} m^2 and resistivity 35×10^{-6} Ω.m. They are all connected in series with an AC source of negligible internal resistance, frequency 50 Hz and the effective value of emf 20 V, then:

(i) The maximum value of the current passing through the circuit is ..... .

(a) 2 A
(b) 2.828 A
(c) 3 A
(d) 3.828 A

(ii) The effective values of the potential difference across the terminals of each of the capacitor and the inductor respectively are ..... .

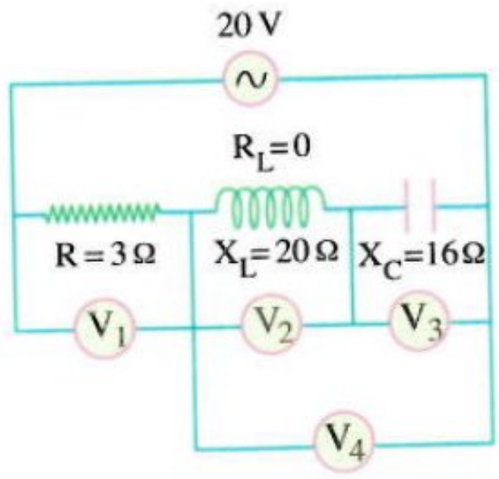
(a) 248.86 V, 226.24 V
(b) 176 V, 160 V
(c) 160 V, 176 V
(d) 226.24 V, 248.86 V

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87 ✱ A coil of self inductance $\frac{7}{220}$ H, ohmic resistance 4 Ω is connected in series to a capacitor of capacitive reactance 5 Ω and a variable ohmic resistance (R), all are connected to an AC electric source of emf 13 V and frequency 50 Hz whose internal resistance is negligible. If the effective value of the current intensity passing in the coil musn't exceed 1 A, then the minimum value for the ohmic resistance (R) that can be used to guarantee the safety of the circuit equals ..... .

(a) 4 Ω (b) 8 Ω (c) 10 Ω (d) 12 Ω

88 ✱ From the circuit illustrated in the figure:



(i) The total impedance of the circuit is ..... .

(a) 5 Ω (b) 7 Ω

(c) 24 Ω (d) 39 Ω

(ii) The value of the current passing in the circuit equals ..... .

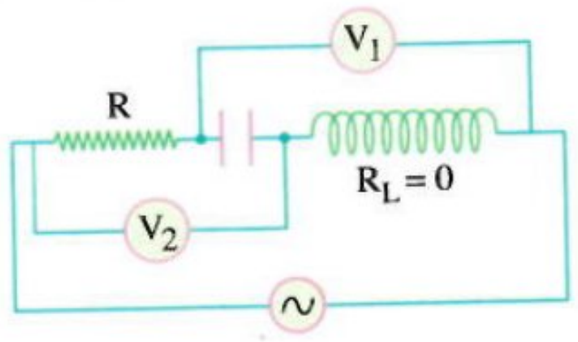
(a) 0.51 A (b) 1 A

(c) 2.85 A (d) 4 A

(iii) The reading of each of the four voltmeters V_1, V_2, V_3, V_4 are ..... .

	V_1	V_2	V_3	V_4
(a)	80 V	12 V	16 V	20 V
(b)	12 V	20 V	20 V	0
(c)	12 V	80 V	64 V	16 V
(d)	64 V	16 V	12 V	144 V

89 In the opposite electrical circuit, if the value of the current through the circuit is 1 A and inductive reactance of the coil is 180 Ω and the reading of each of the two voltmeters is 100 V, then:



(i) The capacitive reactance of the capacitor equals ..... .

(a) 60 Ω (b) 80 Ω (c) 180 Ω (d) 280 Ω

(ii) The value of resistance R equals ..... .

(a) 60 Ω (b) 80 Ω (c) 180 Ω (d) 280 Ω

(iii) The phase angle between the total voltage and the current in the circuit is approximately equal to ..... .

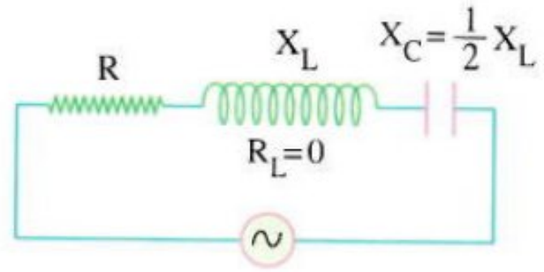
(a) 0° (b) 29° (c) 51° (d) 59°

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90 ✱ In the opposite circuit, if the phase angle between the total voltage and the current is 30°, so when the capacitor is connected to another identical capacitor in parallel, the phase angle between the total voltage and the current (θ) becomes ..... .



(a) 22.65° (b) 36.24° (c) 40.89° (d) 50.92°

91 ✱ An AC source that has a voltage of 400 V and a frequency of $\frac{5000}{\pi}$ Hz is connected in a circuit that contains an inductor of self-inductance 0.2 H, an ohmic resistance of 500 Ω and a capacitor of variable capacitance, so the capacitance which makes the total voltage lags the current vector by an angle of $\frac{\pi}{4}$ equals ..... .

(a) 5×10^{-8} F (b) 4×10^{-8} F (c) 6×10^{-6} F (d) 4×10^{-6} F

92 In figure (1), an RLC, AC circuit, and figure (2) graphically represents the change in each of the total voltage (V) in the circuit and the current (I) flowing through it versus time (t).

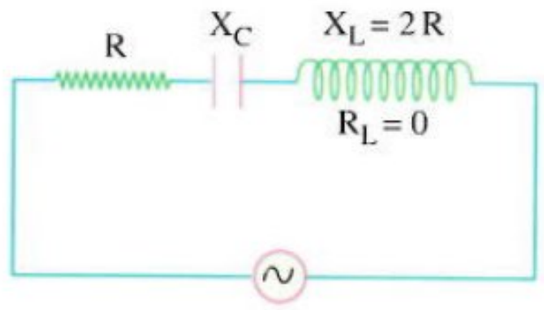


Figure (1)

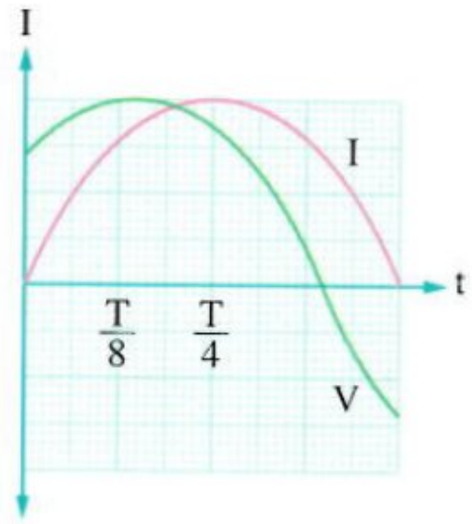
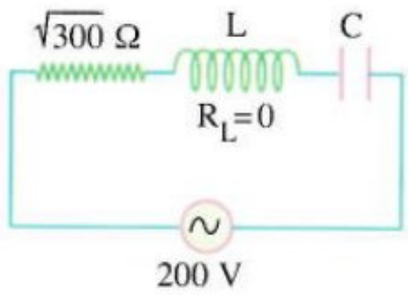


Figure (2)

So, the capacitive reactance (X_C) of the capacitor equals ..... .

(a) $\frac{R}{2}$ (b) R (c) $\frac{3}{2}$ R (d) 3R

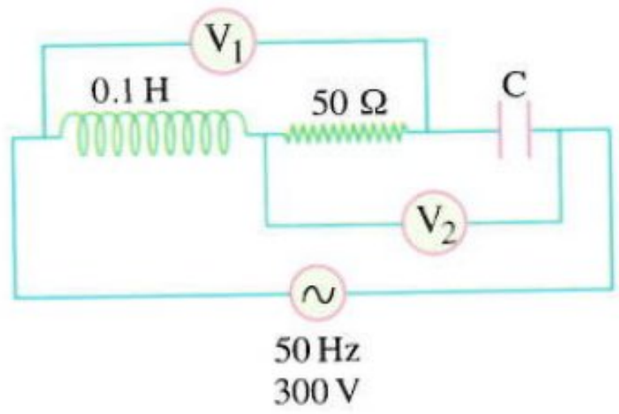
93 In the opposite circuit, if the capacitor only is removed, the total voltage leads the current in phase by an angle of 30° and if the coil only is removed, the total voltage lags the current in phase by an angle of 60°, then the value of the electric current which passes in the opposite electric circuit is approximately ..... .



(a) 3.78 A (b) 7.56 A (c) 9.45 A (d) 18.92 A

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94 ✳ In the opposite electric circuit, if the ratio between the readings of the two voltmeters $\left(\frac{V_1}{V_2}\right)$ is $\frac{1}{2}$, then the capacitance of the capacitor (C) is approximately equal to ..... .

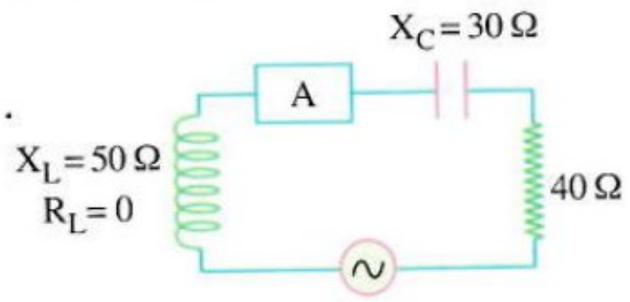


(a) 60 μF (b) 30 μF

(c) 15 μF (d) 7.5 μF

---

95 ✳ In the opposite electric circuit, if the total voltage lags the current by an angle of 45°, then component A is ..... .



(a) an inductor of reactance 20 Ω

(b) an inductor of reactance 80 Ω

(c) a capacitor of reactance 20 Ω

(d) a capacitor of reactance 60 Ω

---

96 ✳ An electric generator of negligible resistance consists of a coil of 500 turns, the cross-sectional area of each is $\frac{7}{11}$ m^2, placed in a uniform magnetic field of flux density 5×10^{-4} T, rotating at frequency of 50 Hz, its terminals are connected to a capacitor of capacitive reactance 110 Ω, an ohmic resistance of 40 Ω and an inductor of inductive reactance 80 Ω, then:

(i) The maximum value of the voltage across the inductor equals ..... .

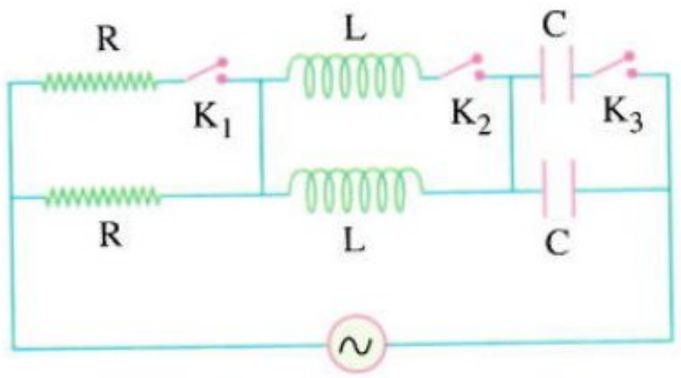
(a) 40 V (b) $40\sqrt{2}$ V (c) 80 V (d) 110 V

(ii) The effective value of the AC current in the circuit equals ..... .

(a) 0.707 A (b) 1 A (c) 1.41 A (d) 2 A

---

97 In the opposite circuit, if $X_C = 2\,X_L = 2\,R$ and the total voltage lags the source current by angle of 45°, which of the switches should be closed so that the phase angle between the total voltage and the source current equals zero?



(a) Switch K_1 only.

(b) Switch K_2 only.

(c) Switch K_3 only.

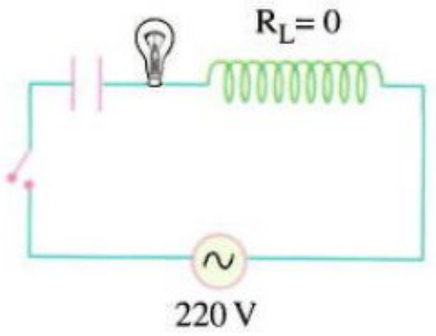
(d) Closing the three switches together.

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98 ✱ In the opposite figure, an electric circuit contains an AC source of emf 220 V, a capacitor of capacitive reactance 800 Ω, a coil of inductive reactance 800 Ω, an electric lamp of resistance 600 Ω and a switch where all are connected in series, then the value of the current through the circuit when:



(i) closing the circuit equals ..... .

(a) $\frac{1}{10}$ A (b) $\frac{4}{10}$ A (c) $\frac{11}{40}$ A (d) $\frac{11}{30}$ A

(ii) removing the capacitor only from the circuit equals ..... .

(a) 0.16 A (b) 0.22 A (c) 0.36 A (d) 0.4 A

(iii) removing the coil only from the circuit equals ..... .

(a) 0.4 A (b) 0.36 A (c) 0.22 A (d) 0.16 A

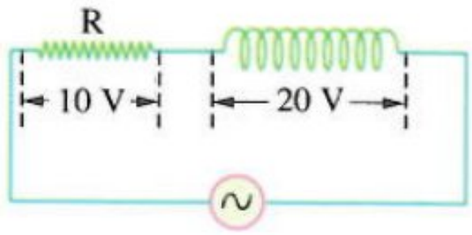
(iv) removing the capacitor and the coil from the circuit equals ..... .

(a) $\frac{11}{30}$ A (b) $\frac{11}{40}$ A (c) $\frac{4}{10}$ A (d) $\frac{3}{10}$ A

Second Essay questions

An electric circuit containing an AC source and two components

1 An ohmic resistor R and an inductor of self inductance L and ohmic resistance R are connected to an AC source, if the voltages across the terminals of the inductor and the resistor are as shown in the opposite figure, **find:**

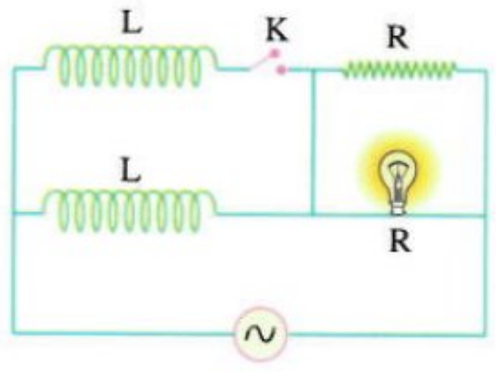


(a) The ratio between the inductive reactance of the inductor and its ohmic resistance.

(b) The potential difference between the terminals of the source.

2 Two similar resistance wires x, y where wire x is double folded before being coiled and wire y is coiled in the form of an inductor and connected with wire x and an AC source in series. **Explain why** the potential difference across the terminals of each of them is different from the other.

3 In the opposite circuit, two adjacent similar coils of negligible ohmic resistance, **what will happen with explanation** to the brightness of the bulb and the phase angle between the total voltage and the current in the circuit at closing the switch K?

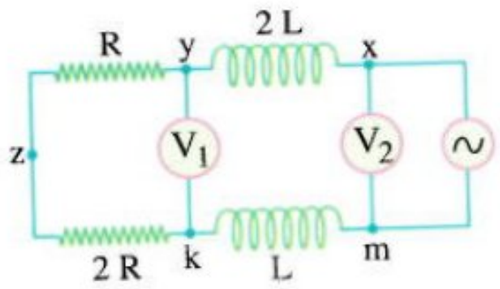


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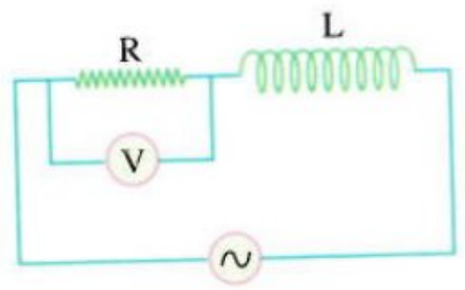
4 In the AC circuit shown in the opposite figure, if the readings of the two voltmeters V_1, V_2 are respectively 30 V, 50 V and the two coils have zero ohmic resistance, **find:**

(a) The potential difference between the two points x, y.

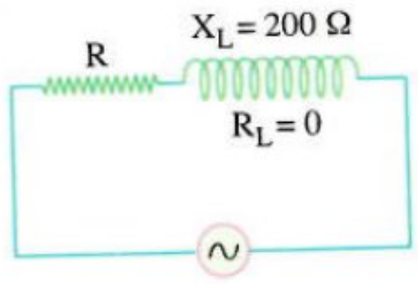
(b) The potential difference between the two points m, z.



5 In the opposite electric circuit, if the turns of the coil are uniformly displaced away from each other, **what will happen to** the reading of the voltmeter?



6 **What happens when** replacing the coil with an ohmic resistance of 200 Ω in the opposite circuit concerning the value of the current through the circuit?

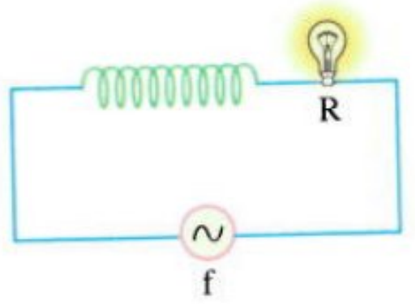


7 In the opposite electric circuit, a lightbulb of resistance R is connected in series with a coil of inductance L and an AC source.

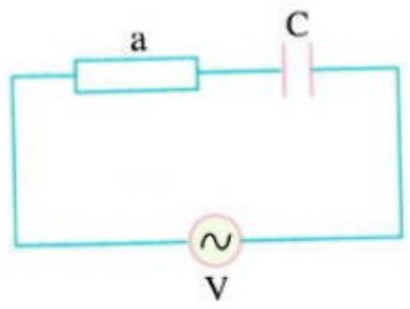
What will happen to the brightness of the bulb when:

(a) increasing the frequency of the source to the double at constant emf?

(b) increasing each of the frequency of the source and its emf to the double?

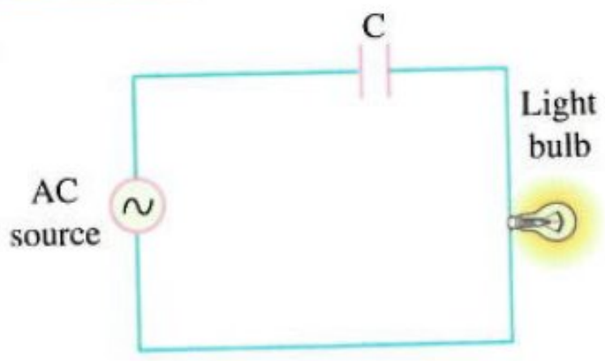


8 In the electric circuit shown in the figure, the capacitive reactance of capacitor (C) equals the ohmic resistance of wire (a), if the wire is drawn uniformly until its length increases to the double, **calculate** the change in the phase angle between the total voltage and the current in the circuit.



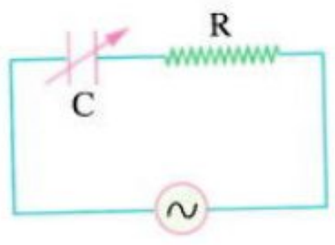
9 In the opposite figure, an AC circuit consists of a capacitor, a glowing bulb, a resistor and an AC source of constant effective voltage value and changeable frequency.

Explain why the glowing of the filament of the bulb decreases when the frequency of the source decreases.



10 In the circuit shown in the opposite figure:

If the phase angle between the current and the total voltage is −30°, **explain how** the capacitance of the capacitor can be changed such that the phase angle becomes −60°

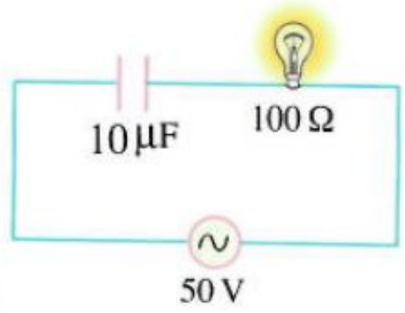


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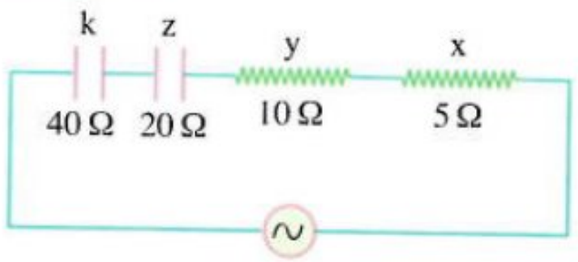
11 In the opposite figure, if the maximum value of the accumulated charge on the plates of the capacitor is $300\sqrt{2}\ \mu C$, **calculate** the consumed power through the bulb.



12 **In the opposite electric circuit, how can you with eliminating only one component from the circuit:**

(a) increase the phase angle between the total voltage and the current to the largest value?

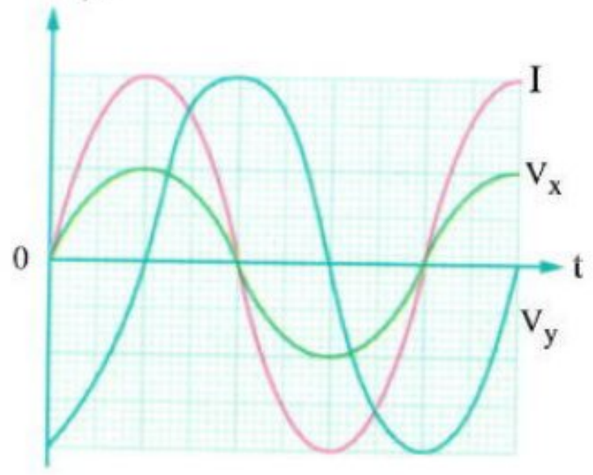
(b) increase the value of the electric current to the largest value?



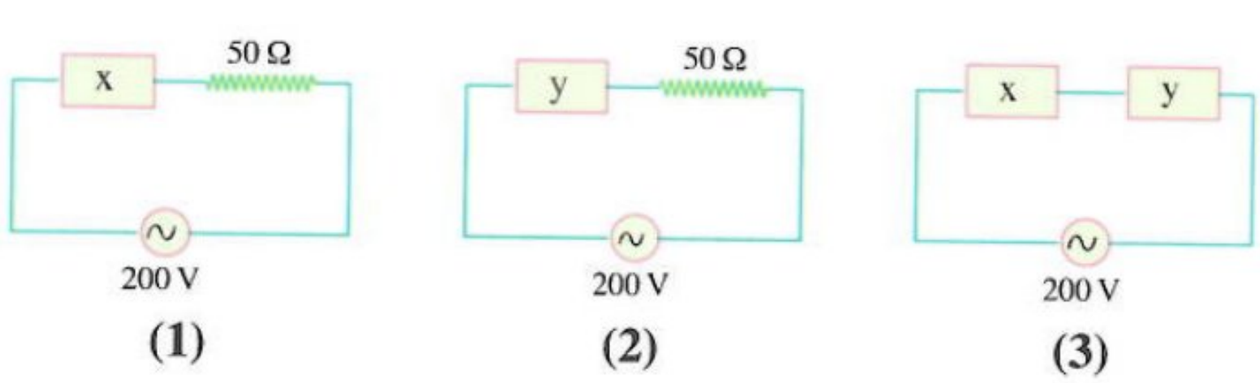
13 AC source of negligible internal resistance is connected to two pure components x, y in series, the opposite graph represents the variation of the electric current and the voltage across each of the two components versus the time :

(a) **Determine the types of** the two components x, y.

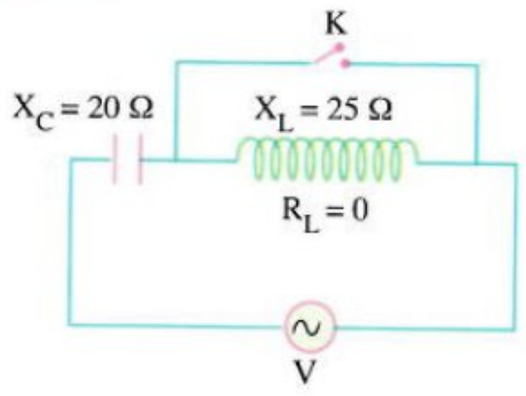
(b) **Find** the phase angle between the total voltage and the current in the circuit.



14 AC circuit consists of an ohmic resistance 50 Ω, a pure component (x) and AC source of voltage 200 V as shown in figure (1), if the phase angle between the total voltage and the current in the circuit is (-45°) and when replacing components (x) by an other pure components (y) as in figure (2), the phase angle between the total voltage and the current becomes (30°) so if the two components x and y are connected together in series to source as in the figure (3), **what** will be the value of the current in the circuit in the last case?

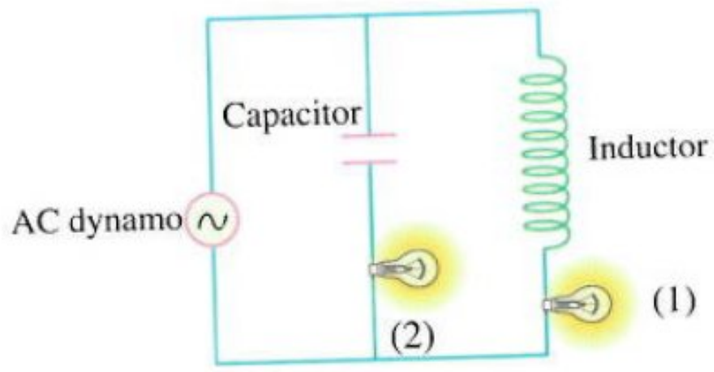


15 In the electric circuit shown in the opposite figure, **what happens to** the value of the electric current through the circuit at closing the switch K?



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16 In the opposite figure, an AC dynamo of constant effective voltage value has a changeable frequency. If the two electric bulbs are similar and the inductive reactance of the coil is equal to the capacitive reactance of the capacitor, then **explain why** by increasing the frequency of rotation of the dynamo coil, the brightness of bulb (2) becomes greater than that of bulb (1).

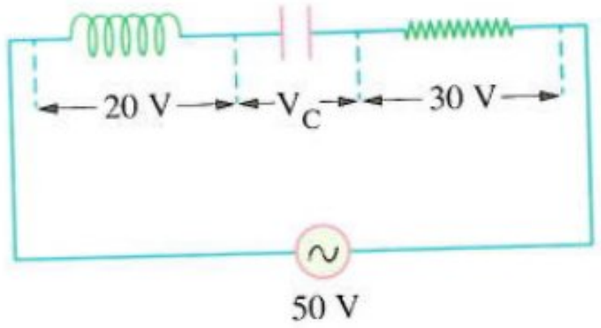


An electric circuit containing an AC source and three components

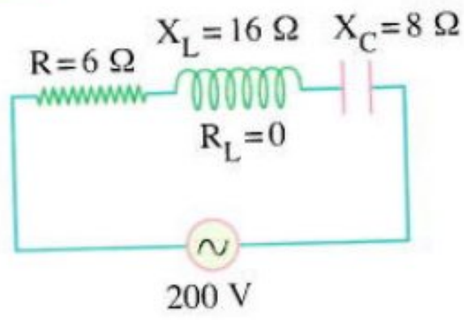
17 The opposite electric circuit represents an RLC circuit connected to an AC source.

(a) **Calculate** the value of the potential difference between the plates of the capacitor (V_C).

(b) **Explain why** the potential difference between the plates of the capacitor (V_C) might be greater than that across the terminals of the source (V).



18 In the opposite RLC circuit, **calculate** the consumed electric power in the circuit.



19 A circuit consists of a capacitor of reactance 30 Ω, a resistor of 44 Ω and a coil of inductive reactance 90 Ω and resistance 36 Ω that are connected in series to an AC source of frequency 60 Hz and voltage 200 V, **calculate:**

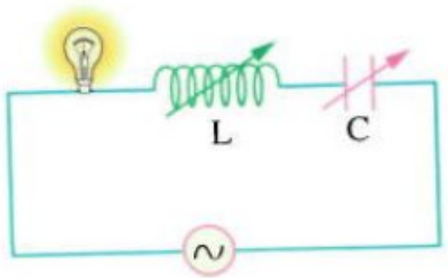
(a) The value of the current passing through the circuit.

(b) The voltage across the coil.

20 In the shown circuit, if $X_L = 2\ X_C$, **what happens to the brightness of the bulb when:**

(a) increasing L to the double.

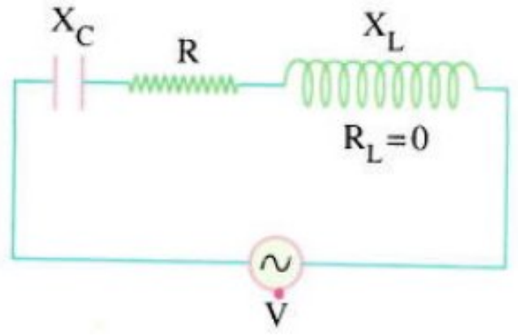
(b) increasing C to the double.



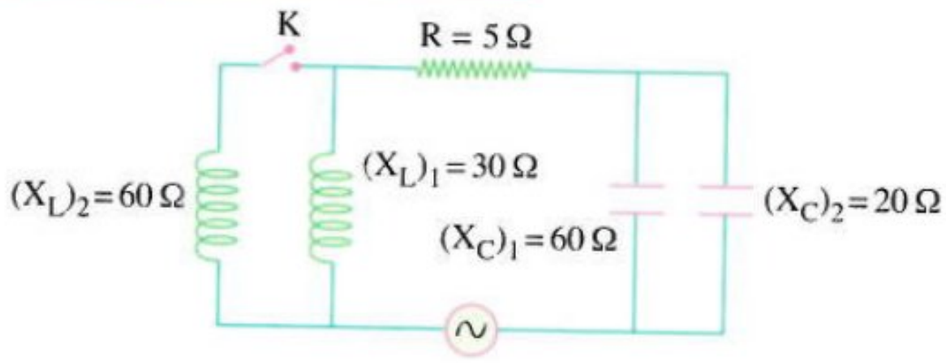
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21 In the electric circuit shown in the figure, if $(X_C < X_L)$, **explain why** the phase angle between the total voltage (V) and the current (I) increases in the circuit after removing the capacitor.



22 In the electric circuit shown in the figure, **what** is the value of the phase angle between the total voltage and the current in the circuit when the switch K is:
(Where : the two coils have negligible ohmic resistance)



(a) opened?

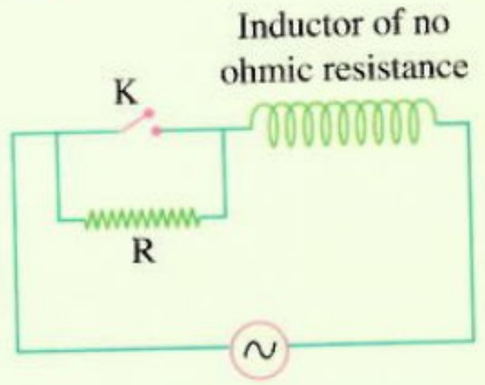
(b) closed?

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Questions of the Previous Years Exams ? on Lesson 2

An electric circuit containing an AC source and two components

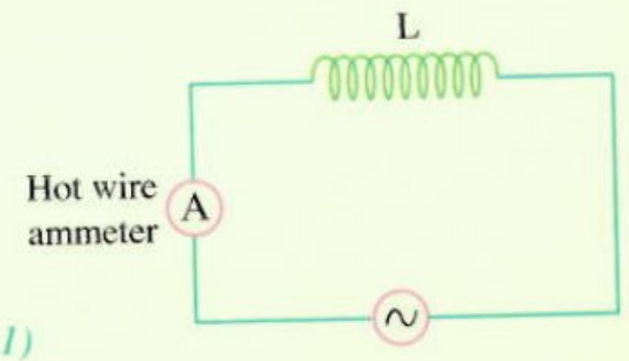
1 In the opposite AC circuit, as the switch (K) is closed the phase angle between the total voltage and current will ..... . *(Exp. May 21)*



(a) not change (b) increase

(c) decrease (d) vanish

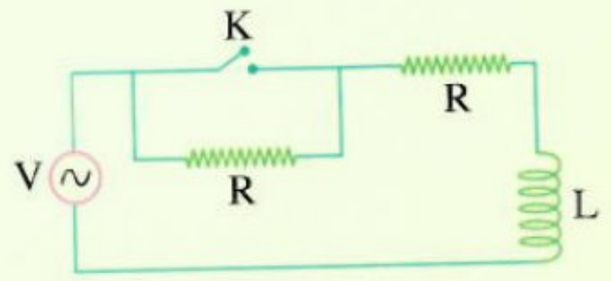
2 The opposite figure shows a series AC circuit containing an AC source of maximum voltage 250 V, an inductor of negligible ohmic resistance and a hot wire ammeter of resistance 12 Ω. If the reading of the ammeter = 10 A, the inductive reactance of the coil = ... *(Exp. May 21)*



(a) 17.67 Ω (b) 12.98 Ω

(c) 21.93 Ω (d) 5.68 Ω

3 The figure illustrates an alternating current circuit, so on closing the key (K), the phase angle between the total voltage (V) and the current (I) ..... . *(Exp. June 21)*

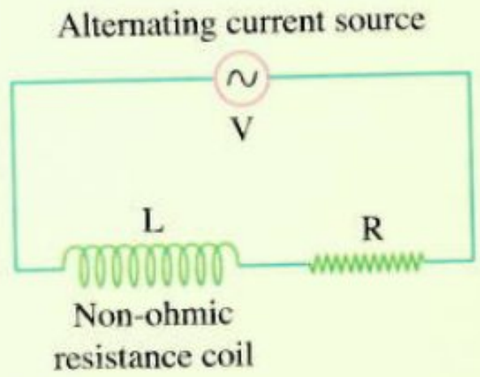


(a) increases (b) decreases

(c) doesn't change (d) becomes zero

4 The opposite figure illustrates an inductive coil of negligible ohmic resistance in an alternating current circuit:

If the source is replaced with another of lower frequency at constant (V), so ..... . *(1st session 21, Practice 25)*



	The inductive reactance of the coil	The phase angle between total voltage and the current
(a)	decreases	increases
(b)	increases	decreases
(c)	decreases	decreases
(d)	increases	increases

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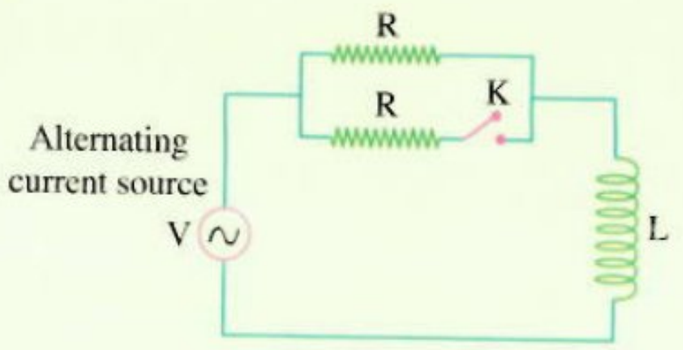
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5 The figure illustrates alternating current circuit: On closing the key (K), the phase angle between total voltage (V) and the current (I) ..... .

(a) decreases (b) doesn't change

(c) increases (d) becomes zero

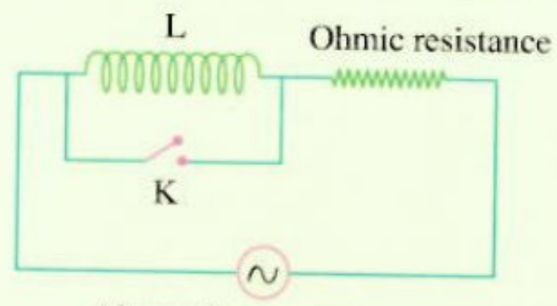


(2nd session 21, Practice 25)

6 An AC circuit contains a non inductive ohmic resistance and a coil of self inductance (L) of negligible ohmic resistance so that the phase angle between the voltage and the current is (θ). When the key (K) is closed, so the phase angle ..... . *(1st session 22)*

(a) becomes zero (b) doesn't change

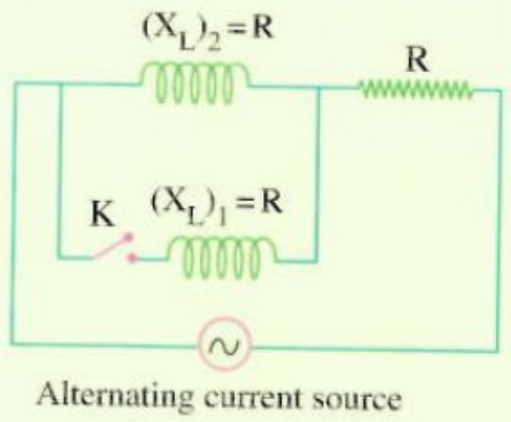
(c) increases (d) decreases but doesn't equal zero



7 An AC circuit contains an ohmic resistance and two inductors of negligible ohmic resistance, such that when the key (K) is open, the phase angle between the total voltage and the current is (θ), so when the key (K) is closed, the phase angle between the total voltage and current ..... . *(2nd session 22)*

(a) increases (b) decreases but doesn't reach zero

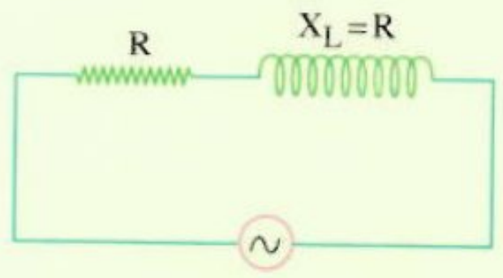
(c) becomes zero (d) doesn't change



8 In the opposite figure, an inductive coil of negligible ohmic resistance, if $\frac{1}{4}$ of the coil is cut and the rest of the coil is connected to the circuit without changing any other factors, the correct choice is ..... . *(1st session 23)*

(a) the phase angle decreases by 8.13° (b) the phase angle decreases by 36.87°

(c) the phase angle decreases by 30.96° (d) the phase angle decreases by 14.04°



9 A straight wire was connected to an AC source in which the effective current intensity was (I), then the wire was wound in the form of a coil and connected to the same source, so the effective current intensity in it becomes ..... . *(Practice 25)*

(a) decreases (b) remains constant

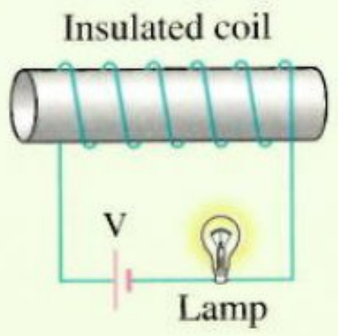
(c) increases (d) the answer cannot be determined

10 In an AC circuit, an inductor whose inductive reactance is 40 Ω and its ohmic resistance is 30 Ω, is connected to an AC source of effective voltage 60 V, so the electrical power consumed in the circuit equals ..... . *(Practice 25)*

(a) 43.2 W (b) 51.4 W (c) 72 W (d) 120 W

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11 An ideal inductor (coil with negligible ohmic resistance) is connected in series with an electrical lamp of ohmic resistance R and a battery of electromotive force V and negligible internal resistance, so the current flowing in the circuit is I and the power consumed in the lamp is P_w and when an alternating current source with voltage V is connected instead of the battery, the current flowing in the circuit becomes 0.75 I, so the power consumed in the lamp and the phase angle between the total voltage and the current in this case respectively are ..... , ... *(Practice 25)*



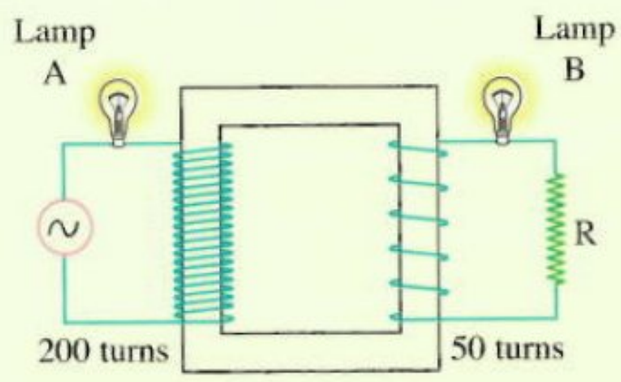
(a) $\frac{9\,P_w}{16}$, 41.41°

(b) $\frac{9\,P_w}{16}$, 36.89°

(c) $\frac{3\,P_w}{4}$, 48.6°

(d) $\frac{3\,P_w}{4}$, 41.41°

12 The figure shows an ideal electrical transformer. When the alternating current source is replaced by a battery whose electromotive force is equal to the effective value of the alternating source voltage, then the brightness of lamp A and that of B, respectively, will ..... . *(Practice 25)*



(a) increase, cease

(b) not change, increase

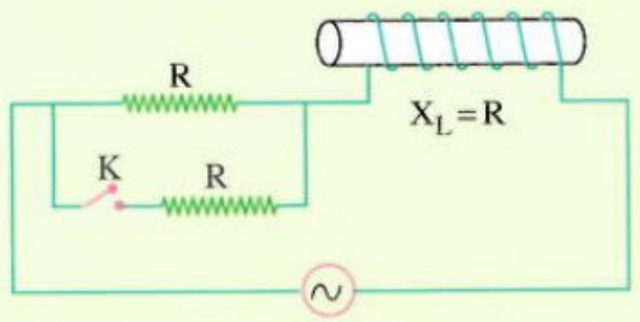
(c) cease, cease

(d) not change, cease

13 The figure represents a part of electric circuit which connected with an alternating current source.

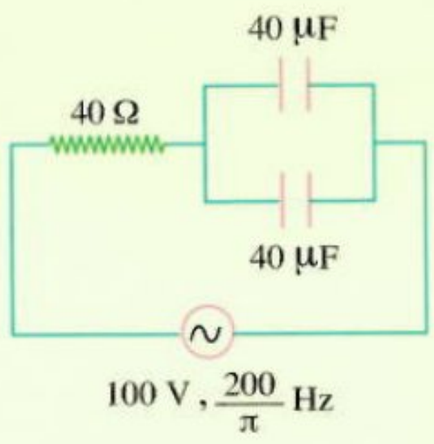
What will happen for the phase angle between the total voltage and the total current at closing key (K)?

Explain your answer.



(Exp. 23)

14 In the shown electric circuit: The phase angle between the total potential difference (V_t) and the current intensity (I) equals ..... . *(1st session 22)*



(a) 38°

(b) 35°

(c) –38°

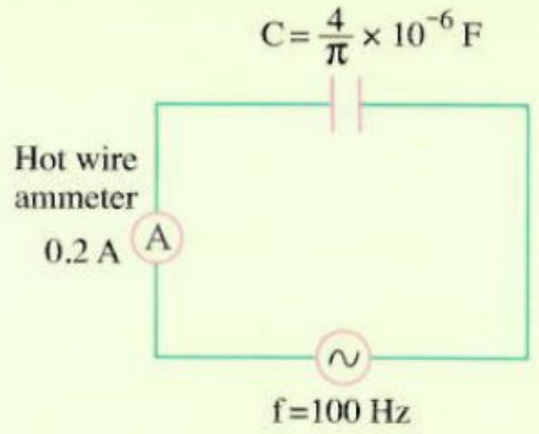
(d) –35°

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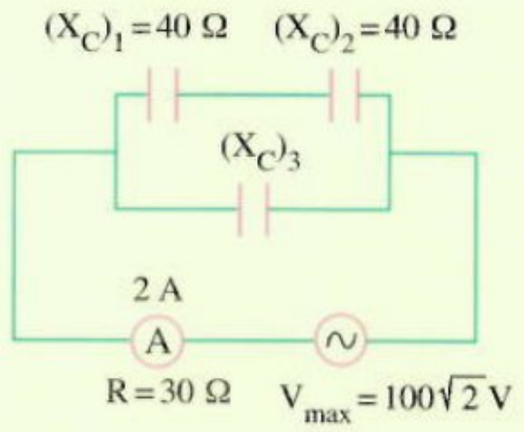
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15 The figure represents an electric circuit that contains a hot wire ammeter of ohmic resistance 50 Ω, an AC source and a capacitor. Using the shown data on the figure, the maximum value of the emf of the AC source equals ..... . *(2nd session 21)*



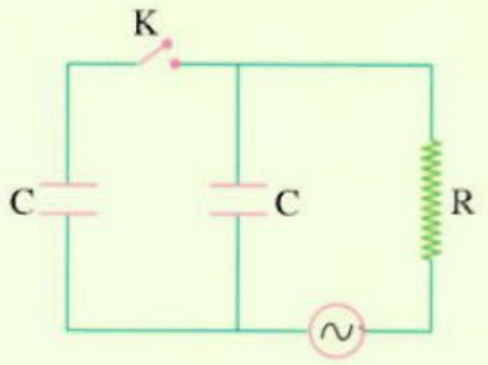
(a) 250.19 V (b) 353.84 V
(c) 194.17 V (d) 318.62 V

16 An AC source produces maximum electromotive force of $100\sqrt{2}$ V and it is connected with three capacitors and a hot wire ammeter as shown in the figure. Using the shown data in the figure, the value of the capacitive reactance $(X_C)_3$ equals ..... . *(2nd session 22)*



(a) 80 Ω (b) 20 Ω
(c) 40 Ω (d) 50 Ω

17 In the shown circuit, the phase angle between the total voltage and the current is (−45°). So, when switch K is closed, the angle becomes ..... . *(Practice 25)*



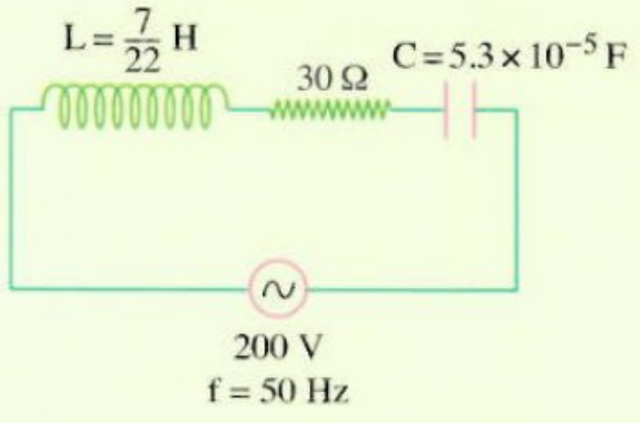
(a) 0° (b) −26.6°
(c) −63.4° (d) −90°

18 An alternating current source of frequency $\frac{100}{\pi}$ Hz and effective voltage 20 V is connected in series with an ohmic resistance of 3 Ω and a capacitor with a capacitance of 1250 μF, so the accumulated charge on one of the capacitor plates when the capacitor is fully charged will be ..... . *(Practice 25)*

(a) 0.020 C (b) 0.025 C (c) 0.028 C (d) 0.035 C

An electric circuit containing an AC source and three components

19 The opposite figure shows an RLC circuit connected with an AC source of emf rated at 200 V and 50 Hz. Using the data on the figure, so the total impedance of the circuit approximately equals ..... . *(Exp. May 21)*

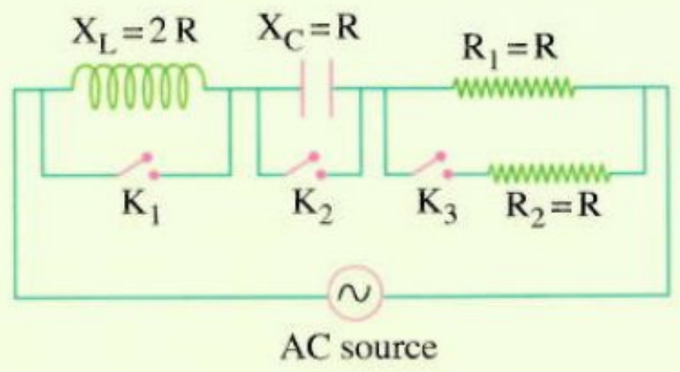


(a) 40 Ω (b) 50 Ω
(c) 100 Ω (d) 30 Ω

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20 The electric circuit shown in the opposite figure contains capacitor, inductive coil of negligible ohmic resistance and two resistances R_1 and R_2. To obtain the maximum consumed power, we have to ..... .

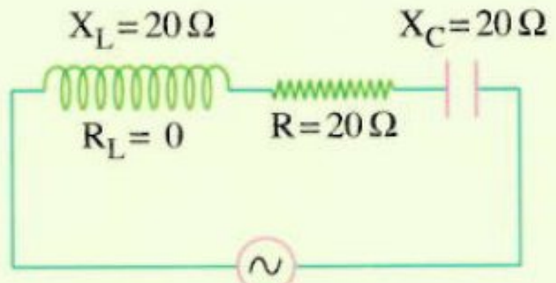
(1st session 24)



(a) open K_1, K_2 and K_3
(b) open K_2 and close K_1 and K_3
(c) close K_2 and K_3 and open K_1
(d) close K_1, K_2 and K_3

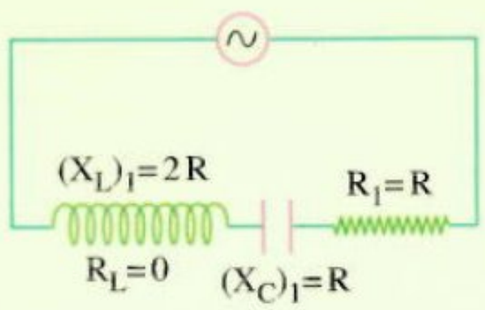
21 From the opposite figure, if the coil is replaced by another coil of the same length, cross-sectional area and material but the number of turns in the second coil is double the number of the first coil, so the ratio between: $\frac{\text{The impedance in the second case}}{\text{The impedance in the first case}}$ = ...

(1st session 24)

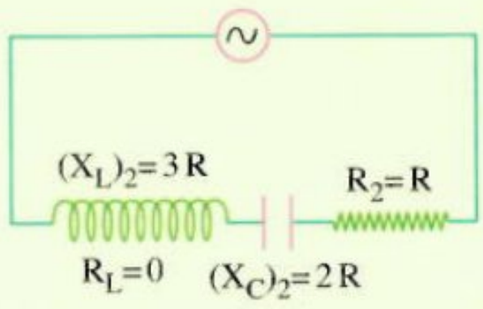


(a) $\sqrt{10}$
(b) $20\sqrt{10}$
(c) $\frac{1}{\sqrt{10}}$
(d) $\frac{1}{20\sqrt{2}}$

22 From the data in the two electric circuits (1), (2),



Circuit (1)



Circuit (2)

the ratio $\left(\frac{Z_1}{Z_2}\right)$ = ...

(2nd session 24)

(a) $\frac{1}{2}$
(b) $\frac{1}{1}$
(c) $\frac{\sqrt{2}}{1}$
(d) $\frac{2}{3}$

23 In a series RLC, AC circuit, the phase angle between the voltage across the coil (V_L) and the voltage across the capacitor (V_C) is ..... .

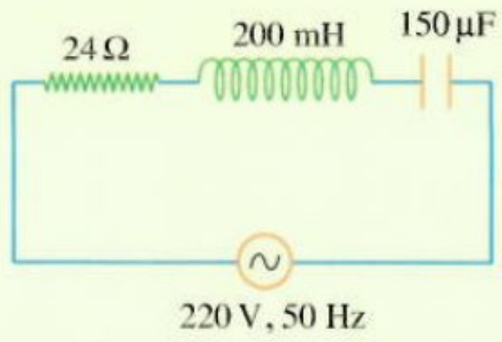
(Practice 25)

(a) 180°
(b) 90°
(c) –90°
(d) 0°

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24 In the circuit shown in the figure, an alternating current source (220 V, 50 Hz) is connected to a capacitor, a resistor and an inductive coil of negligible ohmic resistance in series. What are the values of the total impedance in the circuit and the phase angle between the total voltage and current? (Given that: $\pi = 3.14$) *(2nd session 25)*



	Impedance	Phase angle
(a)	24 Ω	0°
(b)	21.22 Ω	74.21°
(c)	48 Ω	60°
(d)	48 Ω	30.2°

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